

Final Exam Study Guide

Everything from **Week 8 forward** — blood, heart, lungs, kidneys, reproduction, digestion, and the musculoskeletal/integumentary systems — condensed into the mechanisms, comparisons, and lab signatures the exam actually tests.

Dr. Vidya Aggrawal

Weeks 8 – 14

7 systems + integration

120+ practice questions

Cheat sheets & 40-Q rehearsal exam

HOW THIS EXAM THINKS

Nearly every question in Weeks 8–14 is one of five shapes. Learn the shapes and unfamiliar questions stop being unfamiliar.

1. **Mechanism:** "Which process explains..." — they want the *step*, not the disease name.
2. **Discriminator:** two look-alike conditions, one clue separates them (B₁₂ vs folate → neuro signs; OA vs RA → symmetry + stiffness duration).
3. **Lab signature:** a pattern of arrows identifies the disorder (↓platelets + ↑PT + ↑aPTT + ↑D-dimer = DIC).
4. **Consequence:** given the lesion, predict the downstream effect (CKD → no EPO → anemia).
5. **Red flag:** which finding means *act now* (pain out of proportion, tracheal deviation, absent cremasteric reflex).

Week 8

Blood & coagulation — heaviest lab-signature content

Week 9

Heart — heaviest mechanism/compensation V/Q are guaranteed points

Week 10

Lungs — ABGs and V/Q are guaranteed points

Week 11

Kidney — AKI categories, CKD consequences

Week 12

Reproductive — PCOS, BPH vs cancer, emergencies

Week 13

GI — Crohn vs UC, cirrhosis cascade

Week 14

MSK/skin — arthritis trio, burns, rashes

Hematologic Function

Blood as an organ: how it carries oxygen, how it stops bleeding, and the four numbers that decode almost every coagulopathy.

1 Blood, Hemoglobin & the EPO Loop

Blood is ~55% **plasma** and ~45% **formed elements** (RBCs, WBCs, platelets). The hematocrit is the RBC fraction — and it is the number that rises in dehydration and polycythemia and falls in anemia and fluid overload.

Plasma proteins — why albumin matters

Albumin is the main contributor to **plasma oncotic (colloid osmotic) pressure** — the pull that holds fluid inside the vessel. Lose albumin and fluid leaks into tissue.

CROSS-WEEK THREAD Low albumin shows up three more times this exam: **cirrhosis** (liver can't synthesize it — Week 13), **nephrotic syndrome** (kidney spills it — Week 11), and **burns** (capillary leak — Week 14). All three produce the same result: **edema**.

The red cell and hemoglobin

- **Biconcave disc** → large surface area for gas exchange and flexibility to squeeze through capillaries. Lose the shape (sickling, spherocytosis) and cells get trapped and destroyed.
- **No nucleus, no mitochondria** → maximum room for hemoglobin, and the cell can't repair itself → ~120-day lifespan.
- **Hemoglobin** = 4 globin chains (2 alpha, 2 beta in HbA), each with a **heme** ring holding Fe^{2+} . Each Hgb carries **4 O₂**.

Key idea — the oxygen content equation: almost all oxygen in blood is **bound to hemoglobin**, not dissolved. That is why an anemic patient can have a **normal PaO₂** and **normal SaO₂** and still be tissue-hypoxic — the saturation percentage is fine, there is simply not enough hemoglobin to saturate.

The erythropoietin (EPO) feedback loop

Tissue hypoxia → **kidney** peritubular cells sense it → release **EPO** → red marrow → ↑ erythropoiesis → ↑ RBCs → oxygen delivery restored → EPO falls.

Break the loop at any point and you get a predictable anemia or polycythemia:

Break point	Result	Classic setting
Kidney can't make EPO	↓ RBC → normocytic anemia	Chronic kidney disease (Week 11)
Chronic hypoxia drives EPO up	↑ RBC → secondary polycythemia	COPD, high altitude, heavy smoking, cyanotic heart disease
Marrow can't respond	↓ all lines → aplastic anemia / pancytopenia	Chemo, radiation, marrow replacement by leukemia
Raw materials missing	Anemia of a specific size	Iron (microcytic), B₁₂/folate (macrocytic)

Required raw materials

Nutrient	Job	Deficiency picture
Iron	Sits in heme, binds O ₂	Microcytic hypochromic — small, pale cells
Vitamin B₁₂	DNA synthesis for maturation; needs intrinsic factor from gastric parietal cells, absorbed in the terminal ileum	Macrocytic + neurologic signs (paresthesias, ataxia, loss of position sense)
Folate	DNA synthesis	Macrocytic , no neurologic signs

MNEMONIC "B₁₂ has a Brain." Both B₁₂ and folate deficiency make big cells; only B₁₂ damages nerves. That single clue answers the question every time.

PRACTICE

Q1. A patient with end-stage kidney disease has a hemoglobin of 8.2 g/dL with normal MCV. Which mechanism best explains this anemia?

- A. Reduced erythropoietin production by the diseased kidneys
- B. Iron deficiency from poor intake
- C. Antibody-mediated red cell destruction
- D. Defective hemoglobin chain synthesis

A is correct. The kidney is the EPO source. When nephron mass is lost, EPO falls and the marrow is never told to produce — cells made are normal in size, so the anemia is **normocytic normochromic**.

PRACTICE

Q2. Which finding distinguishes vitamin B₁₂ deficiency from folate deficiency?

- A. Numbness, tingling, and unsteady gait
- B. Elevated mean corpuscular volume
- C. Fatigue and pallor
- D. Reduced hemoglobin concentration

A is correct. Both cause macrocytosis, fatigue, and pallor. B₁₂ is additionally required for myelin maintenance, so only B₁₂ deficiency produces **neurologic** deficits.

PRACTICE

Q3. A patient with severe anemia has SaO_2 97% and PaO_2 95 mmHg yet reports dyspnea and fatigue. Which explains the symptoms?

- A. Reduced total hemoglobin lowers oxygen-carrying capacity despite normal saturation
- B. The oxygen in plasma is not being released
- C. Alveolar diffusion is impaired
- D. The patient is hyperventilating

A is correct. Saturation measures the **percentage** of available hemoglobin carrying oxygen — not how much hemoglobin exists. Fewer carriers at 97% full still delivers less total oxygen.

2 Hemostasis & the Coagulation Labs

Primary hemostasis — the platelet plug (seconds to minutes)

1. **Vasoconstriction** of the injured vessel.
2. **Adhesion** — platelets stick to exposed collagen via **von Willebrand factor**.
3. **Activation** — platelets change shape and release ADP and thromboxane A₂.
4. **Aggregation** — platelets bind one another → a soft, temporary plug.

Secondary hemostasis — the fibrin mesh (minutes)

The coagulation cascade converts **prothrombin** → **thrombin**, and thrombin converts **fibrinogen** → **fibrin**, which weaves through the platelet plug and turns it into a stable clot.

Pathway	Trigger	Lab that measures it	Memory hook
Extrinsic	Tissue factor from damaged tissue (fast)	PT / INR	PT = Play Tennis outside (extrinsic)
Intrinsic	Contact with exposed collagen (slower)	aPTT	PTT = Play Table Tennis inside (intrinsic)
Common	Both converge at factor X	Both PT and aPTT prolong	X = the crossroads

Key idea — the four numbers. Nearly every bleeding/clotting question is solved with: **platelet count** (primary hemostasis), **PT/INR** (extrinsic + liver synthesis), **aPTT** (intrinsic), and **fibrinogen/D-dimer** (consumption + breakdown). Learn the *pattern*, not the disease list.

Pearl — the liver is the coagulation factory. Most clotting factors are made in the liver, and several require **vitamin K**. That is why liver failure (Week 13) and vitamin K deficiency both **prolong the PT/INR** — the liver question and the coagulation question are the same question.

PRACTICE

Q1. Which event occurs during *primary* hemostasis?

- A. Platelets adhere to exposed collagen and aggregate into a plug
- B. Fibrinogen is converted to fibrin
- C. Prothrombin becomes thrombin
- D. Plasmin dissolves the clot

A is correct. Primary hemostasis is the **platelet** phase: vasoconstriction, adhesion (via vWF), activation, aggregation. Fibrin formation is **secondary** hemostasis.

PRACTICE

Q2. A boy with hemophilia A has a prolonged aPTT and a normal PT and platelet count. Which deficiency explains this?

- A. Factor VIII — an intrinsic-pathway factor
- B. Factor VII — an extrinsic-pathway factor
- C. Platelet number
- D. Fibrinogen

A is correct. Hemophilia A = **factor VIII** deficiency, an intrinsic factor, so only the **aPTT** prolongs. Platelets are normal, so the bleeding is deep (joints, muscle) rather than petechial.

PRACTICE

Q3. Which pattern of bleeding suggests a *platelet* problem rather than a clotting-factor problem?

- A. Petechiae, purpura, mucosal bleeding, and prolonged oozing from small cuts
- B. Deep muscle hematomas and hemarthrosis
- C. Delayed rebleeding hours after injury
- D. Isolated prolonged aPTT

A is correct. Platelets seal small vessels, so their failure produces **superficial, immediate** bleeding — petechiae and mucosal ooze. Factor deficiencies produce **deep, delayed** bleeding into joints and muscle.

3 Anemia — the MCV Framework

Every anemia produces the same **general** symptoms because the problem is the same: not enough oxygen delivered.

Key idea — general anemia symptoms and why: fatigue and weakness (tissue hypoxia), **pallor** (less hemoglobin pigment + shunting away from skin), **dyspnea on exertion** (demand outstrips delivery), **tachycardia** (raise cardiac output to compensate), **dizziness/headache** (cerebral hypoxia), **cold intolerance** (peripheral shunting).

Classify two ways — know both

By size (MCV) — how labs sort it	By mechanism — how pathophysiology sorts it
Microcytic (<80): iron deficiency, thalassemia, chronic disease, sideroblastic	Blood loss: acute (trauma, GI bleed) or chronic (menorrhagia, occult GI)
Normocytic (80–100): acute blood loss, CKD, aplastic, hemolytic, chronic disease	Decreased production: nutrient lack, marrow failure, no EPO
Macrocytic (>100): B ₁₂ deficiency, folate deficiency, alcohol, liver disease	Increased destruction (hemolysis): sickle cell, autoimmune, G6PD, mechanical

The microcytic discriminator — iron studies

Condition	Serum iron	Ferritin (stores)	TIBC / transferrin	The tell
Iron deficiency	↓	↓ low	↑ high	Empty stores; body raises its "grabbing capacity". Pica, spoon nails, glossitis
Anemia of chronic disease	↓	↑ high/normal	↓ low	Iron is locked in storage , not absent — inflammation sequesters it
Thalassemia	n	n/↑	n	Globin-chain production defect; normal iron; often family/ethnic history

MNEMONIC Ferritin = the pantry. Iron deficiency → the pantry is **empty** (↓ferritin). Chronic disease → the pantry is **full but locked** (↑ferritin, but the cell can't get it).

Hemolytic anemia — recognize the signature

When red cells are destroyed faster than they are made, four things happen at once:

Finding	Why
↑ Reticulocytes	Marrow is working correctly — it is pushing out immature cells to keep up
↑ Indirect (unconjugated) bilirubin → jaundice	Heme breakdown floods the liver's conjugating capacity
↑ LDH	Released from lysed cells
↓ Haptoglobin	Consumed mopping up free hemoglobin

Pearl — the reticulocyte count is the marrow's report card. Anemia with a **high** retic count = destruction or loss (marrow is responding). Anemia with a **low** retic count = production failure (marrow is the problem). This one number splits the entire differential.

PRACTICE

Q1. Labs show MCV 71 fL, low serum ferritin, and elevated TIBC. Which anemia is this?

- A. Iron-deficiency anemia
- B. Anemia of chronic disease
- C. Vitamin B₁₂ deficiency
- D. Aplastic anemia

A is correct. Microcytic with **low ferritin** (empty stores) and **high TIBC** (body increasing its ability to capture iron) is the classic iron-deficiency pattern. Chronic disease shows the opposite: high ferritin, low TIBC.

PRACTICE

Q2. A patient has anemia, jaundice, elevated LDH, elevated reticulocytes, and low haptoglobin. Which process is occurring?

- A. Hemolysis — accelerated red cell destruction
- B. Marrow failure
- C. Iron deficiency
- D. Acute hemorrhage

A is correct. That four-item cluster *is* the hemolysis signature. The high reticulocyte count proves the marrow is healthy and responding — the problem is destruction, not production.

PRACTICE

Q3. Which compensatory change helps maintain oxygen delivery in chronic anemia?

- A. Increased heart rate and stroke volume, with increased 2,3-DPG shifting the curve to release more O₂
- B. Decreased cardiac output to conserve energy
- C. Increased hemoglobin affinity for oxygen
- D. Peripheral vasodilation to the skin

A is correct. The body raises cardiac output, increases **2,3-DPG** (right shift → hemoglobin *releases* oxygen more readily at the tissue), and shunts blood from skin and gut to brain and heart — which is why the patient is pale and tachycardic.

4 Sickle Cell Disease & Polycythemia

Sickle cell — a one-amino-acid problem with whole-body consequences

An autosomal recessive point mutation substitutes **valine for glutamic acid** on the **beta-globin** chain, producing **HbS**.

Deoxygenation → HbS **polymerizes** → cell distorts into a rigid **sickle** → sickled cells **occlude microvessels** → **ischemia + infarction + pain**, and the rigid cells are also destroyed early → **hemolytic anemia**.

Crisis triggers — anything that promotes deoxygenation or sludging

- **Hypoxia** (altitude, flying, respiratory infection)
- **Dehydration** (concentrates HbS)
- **Infection / fever**
- **Cold exposure** (vasoconstriction)
- **Acidosis** and **physical or emotional stress**

RED FLAG — AUTOSPLENECTOMY

Repeated splenic infarction destroys the spleen in childhood. The functionally asplenic patient loses defense against **encapsulated organisms** (*S. pneumoniae*, *H. influenzae*, *N. meningitidis*) → **overwhelming sepsis risk**. Fever in sickle cell disease is an emergency, not a minor illness.

Key idea — crisis management follows the mechanism: hydration (dilutes and unclumps), **oxygen** (reverses the trigger), **analgesia** (the occlusion genuinely infarcts tissue), **warmth**, and **treat infection**. Notice every intervention directly reverses a trigger on the list above.

Polycythemia — too many red cells

	Primary (polycythemia vera)	Secondary
Cause	Marrow neoplasm — autonomous overproduction	Appropriate EPO response to chronic hypoxia
EPO level	↓ low (marrow ignores it)	↑ high
Setting	Older adult; all three cell lines often rise	COPD, high altitude, smoking, cyanotic heart disease, sleep apnea

Shared consequence: ↑ hematocrit → **hyperviscosity** → sluggish flow → **thrombosis, headache, ruddy complexion, hypertension**

PRACTICE

Q1. Which change directly initiates a vaso-occlusive crisis in sickle cell disease?

- A. Deoxygenation causes HbS to polymerize, deforming red cells that then obstruct microvessels
- B. Antibodies coat and destroy red cells
- C. Iron accumulates in the marrow
- D. Platelet counts rise sharply

A is correct. The sequence is deoxygenation → HbS polymerization → sickling → microvascular occlusion → ischemic pain. Every trigger on the list works by promoting deoxygenation or sludging.

PRACTICE

Q2. A long-time smoker with COPD has hematocrit 58% and elevated erythropoietin. Which explains the polycythemia?

- A. Chronic hypoxemia stimulates renal EPO release, increasing red cell production
- B. A marrow neoplasm producing cells autonomously
- C. Dehydration alone
- D. Loss of plasma volume into tissue

A is correct. This is **secondary** polycythemia — an appropriate response to low oxygen. The **high EPO** is the discriminator; in polycythemia vera EPO is **low** because the marrow is producing without being asked.

PRACTICE

Q3. Why is fever in a child with sickle cell disease treated as an emergency?

- A. Repeated splenic infarction leaves the child functionally asplenic and vulnerable to encapsulated-organism sepsis
- B. Fever directly ruptures red cells
- C. Fever causes iron overload
- D. Fever prolongs the PT/INR

A is correct. Autosplenectomy removes a key filter for encapsulated bacteria, so infection can become overwhelming sepsis rapidly. (Infection is also itself a crisis trigger — a double hit.)

5 Bleeding Disorders, Clotting Disorders & DIC

Bleeding — too little clot

Disorder	Defect	Labs	Clue
Thrombocytopenia / ITP	Too few platelets (immune destruction in ITP; often post-viral in children)	↓ platelets; PT and aPTT normal	Petechiae, purpura, mucosal bleeding
Hemophilia A	Factor VIII deficiency; X-linked recessive (affects males)	↑ aPTT; normal PT and platelets	Hemarthrosis , deep muscle bleeds
von Willebrand disease	vWF deficiency → platelets can't adhere (also carries factor VIII)	Normal or ↑ aPTT; platelet <i>count</i> normal but function impaired	Most common inherited bleeding disorder; heavy menses, epistaxis
Liver disease / vitamin K deficiency	Factors not synthesized	↑ PT/INR first	Easy bruising with jaundice or malabsorption

Clotting — too much clot: Virchow's triad

VIRCHOW'S TRIAD 1. **Stasis** of blood flow (immobility, long flights, atrial fibrillation, heart failure) · 2. **Endothelial injury** (trauma, surgery, indwelling lines, smoking, inflammation) · 3. **Hypercoagulability** (cancer, pregnancy, estrogen, inherited thrombophilia, dehydration). **Two or more present = high DVT risk.**

CROSS-WEEK THREAD Virchow's triad is a Week 8 concept that produces a Week 9 problem (**DVT**) and a Week 10 emergency (**pulmonary embolism**). Post-operative, immobile, dehydrated patients hit all three legs at once.

DIC — clotting and bleeding at the same time

Massive trigger (sepsis, obstetric catastrophe, major trauma, burns, cancer) → **widespread microvascular clotting** → platelets and clotting factors are **consumed** → nothing left to clot with → **diffuse bleeding**, while the microclots simultaneously cause **organ ischemia**.

Key idea — DIC is a paradox with one explanation: the bleeding is *caused by* the clotting. This is why the answer to "why is this patient oozing from every IV site?" is **consumption of clotting factors**, not a factor deficiency the patient was born with.

Lab	DIC	Reasoning
Platelets	↓↓	Consumed in microclots
PT / INR	↑	Factors consumed
aPTT	↑	Factors consumed
Fibrinogen	↓	Used up making fibrin
D-dimer / FDP	↑↑	The confirming clue — proof that clots formed <i>and</i> are being broken down

Pearl — treating DIC: the only definitive treatment is **treat the underlying trigger** (usually sepsis). Support with blood products as needed. If you only replace products without stopping the trigger, you are refueling the fire.

PRACTICE

Q1. A septic patient bleeds from IV sites and has ↓platelets, ↑PT, ↑aPTT, ↓fibrinogen, and markedly ↑D-dimer. Which mechanism explains the bleeding?

- A. Widespread clotting consumed platelets and clotting factors
- B. Inherited factor VIII deficiency
- C. Immune destruction of platelets only
- D. Vitamin K deficiency

A is correct. This is **DIC**. The elevated D-dimer proves clots formed and lysed; the low fibrinogen and low platelets prove they were consumed. Inherited disorders would not derange all four numbers simultaneously.

PRACTICE

Q2. Which combination best fits Virchow's triad in a postoperative patient?

- A. Immobility (stasis), surgical vessel trauma (endothelial injury), and postoperative hypercoagulability
- B. Hypertension, hyperlipidemia, and smoking
- C. Fever, leukocytosis, and tachycardia
- D. Anemia, thrombocytopenia, and neutropenia

A is correct. Stasis + endothelial injury + hypercoagulability are the three legs. Surgery supplies all three at once, which is why early ambulation and prophylaxis are standard.

PRACTICE

Q3. A woman reports lifelong heavy menstrual bleeding and frequent nosebleeds; platelet count is normal. Which disorder is most likely?

- A. von Willebrand disease — impaired platelet adhesion
- B. Hemophilia A
- C. Disseminated intravascular coagulation
- D. Polycythemia vera

A is correct. vWD is the **most common inherited** bleeding disorder, affects both sexes, and produces **mucosal** bleeding because platelets cannot adhere — even though the platelet *count* is normal.

6 Leukocyte Disorders & Transfusion Reactions

Leukocyte disorders

Disorder	Mechanism	Key features
Leukemia	Malignant blasts proliferate in marrow and crowd out normal production	The triad follows automatically: anemia (fatigue, pallor), thrombocytopenia (bruising, bleeding), neutropenia (infections) — plus bone pain, hepatosplenomegaly
Lymphoma	Malignancy of lymphoid tissue	Painless lymphadenopathy + "B symptoms" (fever, night sweats, weight loss). Hodgkin = Reed-Sternberg cells , orderly spread; non-Hodgkin = scattered
Neutropenia	Too few neutrophils (chemo, marrow failure, drugs)	Fever may be the only sign of infection — without neutrophils there is no pus and minimal inflammation
Infectious mononucleosis	Epstein-Barr virus infects B cells	Fatigue, pharyngitis, lymphadenopathy, splenomegaly → avoid contact sports (rupture risk)

Key idea — why leukemia causes bleeding *and* infection *and* fatigue: the marrow is a single factory with limited space. Blasts filling it displace **all three** normal lines at once. One lesion, three symptom clusters.

Transfusion reactions

Reaction	Mechanism	Presentation	Timing
Acute hemolytic	ABO incompatibility — recipient antibodies destroy donor cells	Fever, chills, flank/back pain, dark urine, hypotension ; can progress to DIC and AKI	Within minutes
Febrile nonhemolytic	Reaction to donor leukocytes/cytokines	Fever, chills — no hemolysis	Most common reaction
Allergic / anaphylactic	Reaction to donor plasma proteins	Urticaria, itching → wheeze, angioedema, shock	Minutes
TACO (circulatory overload)	Volume infused faster than the heart can handle	Dyspnea, crackles, JVD, hypertension	During/soon after
TRALI	Antibody-mediated lung injury	Acute hypoxemia, bilateral infiltrates, usually normal BP or hypotension	Within 6 hours

RED FLAG — FIRST ACTION

Any suspected transfusion reaction: **stop the transfusion immediately**, keep the IV open with normal saline, and assess. Fever + flank pain + dark urine during a transfusion = **acute hemolytic reaction** until proven otherwise.

Pearl — TACO vs TRALI: both cause dyspnea during transfusion. **TACO is a volume problem** — hypertension, JVD, responds to diuresis. **TRALI is a lung-injury problem** — hypoxemia with normal or low BP and no volume overload.

Pediatric hematology quick hits

- **Physiologic anemia of infancy** — normal nadir around 2–3 months as fetal RBCs are replaced; self-limited.
- **Hemolytic disease of the newborn** — Rh-negative mother sensitized to an Rh-positive fetus; maternal antibodies cross the placenta and hemolyze fetal cells → anemia + jaundice.
- **Iron-deficiency anemia** — peak in toddlers with excessive cow's milk intake.
- **Childhood ITP** — often follows a viral illness; abrupt petechiae in an otherwise well child; usually self-limited.

PRACTICE

Q1. Fifteen minutes into a transfusion a patient develops fever, chills, flank pain, and dark urine. Which reaction is occurring?

- A. Acute hemolytic reaction from ABO incompatibility
- B. Febrile nonhemolytic reaction
- C. Transfusion-associated circulatory overload
- D. Mild allergic reaction

A is correct. Flank pain and **hemoglobinuria** (dark urine) indicate intravascular hemolysis — not just a febrile reaction. Stop the transfusion immediately; this can progress to AKI and DIC.

PRACTICE

Q2. Why does acute leukemia cause infection, bleeding, and fatigue simultaneously?

- A. Blast cells crowd the marrow and suppress production of neutrophils, platelets, and red cells
- B. Blasts directly consume clotting factors
- C. The spleen destroys all cell lines
- D. The kidney stops producing erythropoietin

A is correct. Marrow space is finite. Blast proliferation displaces all three normal lines, producing neutropenia (infection), thrombocytopenia (bleeding), and anemia (fatigue) at once.

PRACTICE

Q3. Which finding most suggests Hodgkin rather than non-Hodgkin lymphoma?

- A. Reed-Sternberg cells on biopsy with orderly node-to-node spread
- B. Painless lymphadenopathy
- C. Night sweats and weight loss
- D. Fatigue

A is correct. Painless nodes and B symptoms occur in both. The **Reed-Sternberg cell** is the defining histologic feature of Hodgkin lymphoma.

WEEK 9

Cardiovascular Function

Pump, pipes, and perfusion — heart failure, ischemia, the compensations that save you and then kill you, and congenital defects.

7 Cardiac Output, Conduction & Coronary Perfusion

Cardiac Output = Heart Rate × Stroke Volume | Stroke Volume depends on **Preload**, **Afterload**, and **Contractility**

Determinant	Definition	Raised by	Effect on stroke volume
Preload	Volume/stretch in the ventricle at end of diastole — "how full"	Fluid overload, venous return, sodium/water retention, regurgitation	Frank-Starling: more stretch → stronger contraction — <i>up to a point</i> , then the overstretched ventricle fails
Afterload	Resistance the ventricle must overcome to eject — "how hard to push"	Hypertension, aortic stenosis, vasoconstriction, pulmonary hypertension (right side)	↑ afterload → ↓ stroke volume and ↑ myocardial oxygen demand
Contractility	Intrinsic force of contraction	Sympathetic stimulation, calcium, inotropes	↑ contractility → ↑ stroke volume. Lost in ischemia, infarction, cardiomyopathy, acidosis

Key idea — the Frank-Starling curve explains heart failure treatment. A normal heart pumps harder when stretched. A failing heart is already on the flat/descending part of the curve, so **more volume makes it worse**. That is precisely why we give **diuretics** (reduce preload) and **vasodilators** (reduce afterload) rather than fluids.

Conduction — the electrical chain of command

Pacemaker	Intrinsic rate	Note
SA node	60–100 /min	The normal pacemaker — the fastest site wins
AV node	40–60 /min	Deliberately delays the impulse so the atria finish filling the ventricles ("atrial kick")
Bundle branches / Purkinje	20–40 /min	Last-resort escape rhythm — too slow to perfuse well

Pearl — atrial fibrillation ties Weeks 8 and 9 together. In A-fib the atria quiver instead of contracting; you lose the atrial kick (↓ cardiac output) *and* blood pools in the atrium — **stasis**, the first leg of Virchow's triad → mural thrombus → **embolic stroke**. This is why A-fib patients are anticoagulated.

Coronary perfusion — a timing problem

Key idea: the coronary arteries fill during **diastole**, when the myocardium is relaxed. Therefore **tachycardia shortens diastole and reduces coronary filling at exactly the moment oxygen demand is highest** — the core reason exertion or tachyarrhythmia precipitates angina.

Capillary exchange — the four causes of edema

Mechanism	Example
↑ Capillary hydrostatic pressure (pushes fluid out)	Heart failure, venous obstruction, fluid overload
↓ Plasma oncotic pressure (less pull holding fluid in)	Low albumin: cirrhosis, nephrotic syndrome, malnutrition
↑ Capillary permeability (leaky wall)	Inflammation, sepsis, burns, allergic reactions
Lymphatic obstruction (no drainage)	Lymphedema after node dissection, filariasis, tumor

PRACTICE

Q1. Which change would most reduce stroke volume in a patient with hypertension?

- A. Increased afterload — the ventricle must overcome higher resistance to eject
- B. Increased preload
- C. Increased contractility
- D. Decreased heart rate

A is correct. Chronic hypertension raises the resistance against which the left ventricle ejects. Stroke volume falls and myocardial oxygen demand rises — over time producing hypertrophy and then failure.

PRACTICE

Q2. Why does sustained tachycardia worsen myocardial ischemia?

- A. It shortens diastole, reducing coronary filling time while increasing oxygen demand
- B. It dilates the coronary arteries excessively
- C. It increases diastolic filling time
- D. It lowers myocardial oxygen consumption

A is correct. Coronaries perfuse during diastole. Faster rates shorten diastole (supply ↓) while raising workload (demand ↑) — a supply-demand mismatch on both sides at once.

PRACTICE

Q3. A patient with cirrhosis develops peripheral edema and ascites with normal cardiac function. Which mechanism is primarily responsible?

- A. Decreased plasma oncotic pressure from low albumin synthesis
- B. Increased capillary permeability
- C. Lymphatic obstruction
- D. Increased cardiac output

A is correct. The failing liver cannot synthesize albumin, so oncotic pull drops and fluid stays in the interstitium. (Portal hypertension adds a hydrostatic component for the ascites specifically.)

8 Heart Failure — the Forward/Backward Framework

Key idea — one framework answers every heart failure question. Ask two questions: **(1) Which side failed?** and **(2) Is the symptom from blood not going forward, or from blood backing up?** Blood backs up *behind* the failing ventricle. Left ventricle → backs up into the **lungs**. Right ventricle → backs up into the **body**.

	LEFT-sided failure	RIGHT-sided failure
Blood backs up into	Pulmonary circulation	Systemic venous circulation
Backward signs	Dyspnea, orthopnea, paroxysmal nocturnal dyspnea, crackles, cough, pink frothy sputum (pulmonary edema), S3	Jugular venous distention, peripheral/dependent edema, hepatomegaly, splenomegaly, ascites, weight gain, anorexia/nausea from gut congestion
Forward signs	Shared: fatigue, weakness, exercise intolerance, cool extremities, oliguria, confusion — not enough output to perfuse tissue	
Most common cause	Hypertension, coronary artery disease/MI	Left-sided heart failure (the #1 cause). From lung disease instead = cor pulmonale

MNEMONIC Left → Lungs. Right → Rest of the body. If you remember only one thing about heart failure, remember this.

Systolic vs diastolic failure

	Systolic (HFrEF)	Diastolic (HFpEF)
Problem	Ventricle can't contract/eject — contractility lost	Ventricle can't relax/fill — stiff, non-compliant wall
Ejection fraction	↓ Reduced (<40%)	Preserved (normal EF, but small filling volume)
Typical cause	MI, dilated cardiomyopathy, chronic volume overload	Long-standing hypertension → hypertrophy, aging, aortic stenosis

Pearl — orthopnea and PND have a mechanical explanation. Lying flat returns pooled leg fluid to the circulation, increasing preload the failing left ventricle cannot handle → pulmonary congestion. That is why patients sleep on pillows (orthopnea) and wake gasping 1–2 hours after lying down (**paroxysmal nocturnal dyspnea**).

Pearl — BNP. B-type natriuretic peptide is released by **stretched ventricular myocardium**. A high BNP says "the ventricle is being stretched by volume" — useful to distinguish cardiac dyspnea from pulmonary dyspnea (Week 10).

Pearl — daily weight is the best early monitor. A gain of ~1 kg (2.2 lb) equals roughly 1 liter of retained fluid. Weight rises *before* crackles and edema appear.

PRACTICE

Q1. A patient has jugular venous distention, hepatomegaly, and pitting ankle edema. Which failure is present, and why these findings?

- A. Right-sided failure — blood backs up into the systemic venous circulation
- B. Left-sided failure — blood backs up into the lungs
- C. Left-sided failure — reduced forward output
- D. Aortic stenosis

A is correct. The right ventricle receives systemic venous return; when it fails, pressure backs up into neck veins, liver, and dependent tissue. Left failure would produce crackles and orthopnea instead.

PRACTICE

Q2. Which mechanism explains paroxysmal nocturnal dyspnea in left heart failure?

- A. Recumbency redistributes peripheral fluid centrally, raising preload and causing pulmonary congestion
- B. Bronchospasm during sleep
- C. Reduced surfactant at night
- D. Decreased venous return when supine

A is correct. Lying flat mobilizes leg fluid into the central circulation. The failing left ventricle cannot handle the added preload, pressure backs into the pulmonary capillaries, and the patient wakes breathless.

PRACTICE

Q3. A patient with long-standing hypertension has dyspnea and crackles but an ejection fraction of 58%. Which type of failure is this?

- A. Diastolic failure — a stiff hypertrophied ventricle that cannot fill adequately
- B. Systolic failure with reduced contractility
- C. Right-sided failure
- D. High-output failure

A is correct. A normal ejection *fraction* with congestive symptoms indicates a **filling** problem. Chronic pressure overload thickens the wall, reducing compliance — the ventricle ejects a normal percentage of a too-small volume.

9 Ischemia, Angina & Myocardial Infarction

The atherosclerosis sequence

Endothelial injury (hypertension, smoking, hyperlipidemia, diabetes) → LDL enters and oxidizes → macrophages engulf it → **foam cells** → **fatty streak** → smooth muscle and fibrous cap form a **plaque** → plaque narrows the lumen (**stable angina**) → **plaque ruptures** → platelets adhere → **thrombus** → sudden occlusion (**ACS / MI**).

Key idea — ischemia is reversible, infarction is not. Ischemia = supply-demand mismatch; restore flow and the cell recovers. Infarction = prolonged ischemia → cell death → **necrosis**, replaced permanently by non-contractile scar. This single distinction drives the urgency of "time is muscle."

	Stable angina	Unstable angina	Myocardial infarction
Trigger	Exertion or emotion (predictable)	Occurs at rest , new, or worsening	Often at rest, sudden
Relief	Rest or nitroglycerin	Not reliably relieved	Not relieved
Duration	< 5–15 min	Longer	> 20–30 min
Cell death	No	No	Yes
Troponin	Normal	Normal — the key discriminator	↑↑ Elevated

Pearl — troponin is the most cardiac-specific marker. It rises within a few hours, peaks around 12–24 hours, and stays elevated for days. Unstable angina and MI can look identical clinically; **troponin is what separates them**, because troponin only leaks when myocytes die.

RED FLAG — ATYPICAL PRESENTATIONS

Women, older adults, and people with diabetes frequently present without classic crushing chest pain — instead fatigue, dyspnea, nausea, indigestion, jaw or back discomfort. Diabetic **autonomic neuropathy** can blunt pain entirely (**silent MI**). Never rule out MI because the pain "doesn't sound cardiac."

Key idea — complications follow the anatomy. Necrotic muscle → **dysrhythmias** (irritable tissue — the leading early cause of death), **heart failure/cardiogenic shock** (lost contractility), **papillary muscle rupture** (acute valve regurgitation), **ventricular wall rupture**, **pericarditis**, and **mural thrombus** → **embolic stroke**.

PRACTICE

Q1. Which finding best distinguishes myocardial infarction from unstable angina?

- A. Elevated troponin indicating myocyte necrosis
- B. Chest pain occurring at rest
- C. Pain lasting more than five minutes
- D. Radiation of pain to the left arm

A is correct. Both present with rest pain that nitroglycerin may not relieve. Only infarction kills myocytes, and only dying myocytes release **troponin**.

PRACTICE

Q2. Which event converts stable atherosclerotic disease into an acute coronary syndrome?

- A. Plaque rupture with platelet adhesion and thrombus formation occluding the vessel
- B. Gradual growth of the fibrous cap
- C. Increased LDL entry into the intima
- D. Coronary artery vasodilation

A is correct. Stable plaques narrow gradually and cause exertional angina. **Rupture** exposes thrombogenic material, triggering rapid thrombosis and abrupt occlusion — which is why an MI can strike without prior symptoms.

PRACTICE

Q3. Why is dysrhythmia the most common early cause of death after MI?

- A. Ischemic and necrotic myocardium becomes electrically unstable, triggering lethal rhythms
- B. The heart valves fail immediately
- C. Troponin release poisons the conduction system
- D. Scar tissue conducts too rapidly

A is correct. Injured myocardium depolarizes abnormally and creates reentry circuits, producing ventricular tachycardia and fibrillation — often within the first hours, before pump failure has time to develop.

10 Compensatory Mechanisms — Helpful, Then Harmful

Key idea — the central irony of heart failure. The body senses low cardiac output and responds as though the patient is **hemorrhaging**: raise the rate, squeeze the vessels, retain salt and water. Those responses are perfect for blood loss and **exactly wrong** for a failing pump — they increase both preload and afterload on a ventricle that already cannot cope. **The compensation becomes the disease.**

Mechanism	What it does	Short-term benefit	Long-term harm
Sympathetic nervous system	↑ heart rate, ↑ contractility, vasoconstriction	Maintains blood pressure and output	↑ myocardial O ₂ demand, shortened diastole → ischemia; ↑ afterload; dysrhythmias
RAAS (renin–angiotensin–aldosterone)	Angiotensin II vasoconstricts; aldosterone retains Na ⁺ and water	Restores circulating volume and pressure	↑ preload and ↑ afterload → congestion, edema, greater workload
ADH (vasopressin)	Water retention	↑ volume	Dilutional hyponatremia, worsening congestion
Ventricular hypertrophy & remodeling	Myocytes thicken/chamber dilates	More force generated initially	Thick muscle outgrows its blood supply → ischemia, stiffness (diastolic failure), fibrosis, dysrhythmia
Natriuretic peptides (ANP/BNP)	Promote sodium and water loss , vasodilate	The one counter-regulatory system — genuinely helpful	Eventually overwhelmed by RAAS

Pearl — this table explains the drug classes. Beta blockers blunt the sympathetic response; ACE inhibitors/ARBs and aldosterone antagonists interrupt RAAS; diuretics unload preload. Every major heart failure drug works by **switching off a compensation that has turned harmful**.

Shock — four categories, one endpoint

Type	Core problem	Examples	Hemodynamic clue
Hypovolemic	Not enough volume	Hemorrhage, burns, dehydration, third-spacing	↓ preload, cool clammy skin, tachycardia, ↓ urine output
Cardiogenic	Pump failure	Massive MI, dysrhythmia, cardiomyopathy	↓ output with congestion — crackles, JVD (volume present but not moving)
Obstructive	Physical block to flow	Tension pneumothorax, cardiac tamponade, massive PE	↓ output despite normal volume and normal pump
Distributive	Massive vasodilation → volume in the wrong place	Septic, anaphylactic, neurogenic	Often warm, flushed skin early; low systemic vascular resistance

Key idea — all shock ends the same way: inadequate tissue perfusion → cells switch to **anaerobic metabolism** → **lactic acid** accumulates → **metabolic acidosis** → cellular injury → multi-organ dysfunction. An unexplained rising lactate with metabolic acidosis is a perfusion emergency.

PRACTICE

Q1. Why does chronic RAAS activation worsen heart failure over time?

- A. Sodium and water retention raise preload while vasoconstriction raises afterload, increasing the failing ventricle's workload
- B. It reduces circulating volume too much
- C. It directly destroys myocytes
- D. It lowers blood pressure excessively

A is correct. RAAS is designed for hypovolemia. In heart failure it loads the ventricle from both ends — more volume to handle and more resistance to eject against.

PRACTICE

Q2. Why does ventricular hypertrophy eventually cause ischemia?

- A. Muscle mass increases faster than coronary blood supply can grow to serve it
- B. Hypertrophy narrows the coronary arteries directly
- C. Thickened muscle needs less oxygen
- D. Hypertrophy increases ejection fraction excessively

A is correct. The thickened wall demands more oxygen while its perfusion does not expand proportionally — the muscle outgrows its blood supply, producing ischemia even without coronary plaque.

PRACTICE

Q3. A trauma patient is hypotensive with distended neck veins, tracheal deviation, and absent breath sounds on one side. Which shock category is this?

- A. Obstructive shock from tension pneumothorax impeding venous return
- B. Hypovolemic shock
- C. Distributive shock
- D. Cardiogenic shock

A is correct. Volume and pump are intact; air under pressure compresses the mediastinum and great veins, physically obstructing venous return. Fluids will not fix it — decompression will.

11 Congenital & Pediatric Cardiac

Fetal circulation and the newborn transition

In utero the **placenta** is the gas-exchange organ and the lungs are fluid-filled and high-resistance, so three shunts bypass them:

Structure	Route	Becomes after birth
Ductus venosus	Bypasses the liver → inferior vena cava	Ligamentum venosum
Foramen ovale	Right atrium → left atrium (bypasses lungs)	Fossa ovalis
Ductus arteriosus	Pulmonary artery → aorta (bypasses lungs)	Ligamentum arteriosum

Key idea — what makes the shunts close: the first breaths **expand the lungs and drop pulmonary vascular resistance**; clamping the cord **raises systemic resistance**. Left atrial pressure now exceeds right, snapping the foramen ovale shut, and **rising oxygen tension** constricts the ductus arteriosus. Everything follows from pressure reversal + oxygen.

The organizing question: which way does blood shunt?

	LEFT → RIGHT shunt	RIGHT → LEFT shunt
Why	Left-side pressure is higher, so oxygenated blood recirculates to the lungs	An obstruction or malposition forces deoxygenated blood into systemic circulation
Cyanosis?	Acyanotic — systemic blood is still oxygenated	Cyanotic — deoxygenated blood bypasses the lungs
Consequence	↑ pulmonary blood flow → pulmonary congestion, heart failure signs , poor feeding, poor weight gain, tachypnea, diaphoresis with feeds	Hypoxemia, cyanosis, clubbing, polycythemia (chronic hypoxia → ↑ EPO)
Examples	ASD, VSD, PDA	Tetralogy of Fallot, transposition of the great arteries

MNEMONIC — TETRALOGY OF FALLOT PROVe: Pulmonary stenosis · **R**ight ventricular hypertrophy · **O**verriding aorta · **V**SD. The pulmonary stenosis is what drives blood right-to-left through the VSD → cyanosis.

Pearl — why squatting relieves a "tet spell." Squatting kinks the femoral arteries and **raises systemic vascular resistance**, which makes it harder for blood to shunt right-to-left — so more blood is pushed through the stenotic pulmonary valve to be oxygenated. Older children learn this instinctively.

Defect	Signature finding
Coarctation of the aorta	Hypertension and bounding pulses in the arms; low BP, weak or delayed femoral pulses, and cool legs — an arm-to-leg BP gradient
Patent ductus arteriosus	Continuous "machinery" murmur ; common in premature infants; wide pulse pressure, bounding pulses
VSD	Loud holosystolic murmur ; the most common congenital defect; small ones often close spontaneously
Transposition of the great arteries	Two parallel circulations — survival depends on a mixing point (PFO/PDA/VSD); profound cyanosis at birth

RED FLAG — EISENMENGER SYNDROME

A long-uncorrected **left-to-right** shunt drives chronically high pulmonary blood flow → pulmonary vascular remodeling → **pulmonary hypertension** → right-side pressure eventually exceeds left → the shunt **reverses to right-to-left** and a previously pink child becomes **cyanotic**. At that point the defect is generally no longer repairable — the reason acyanotic defects still get fixed early.

PRACTICE

Q1. Which change causes the foramen ovale to close after birth?

- A. Lung expansion lowers pulmonary resistance so left atrial pressure exceeds right atrial pressure
- B. Rising carbon dioxide levels
- C. Cord clamping lowers systemic resistance
- D. Increased right atrial pressure

A is correct. Breathing drops pulmonary vascular resistance and cord clamping raises systemic resistance. The pressure gradient reverses and presses the flap of the foramen ovale closed.

PRACTICE

Q2. An infant with tetralogy of Fallot squats during a cyanotic spell. Why does this help?

- A. It increases systemic vascular resistance, reducing right-to-left shunting and forcing more blood through the pulmonary valve
- B. It decreases systemic vascular resistance
- C. It closes the ventricular septal defect
- D. It slows the heart rate directly

A is correct. Raising systemic resistance makes the systemic side "harder to enter," so relatively more blood crosses the stenotic pulmonary outflow and gets oxygenated.

PRACTICE

Q3. A child has upper-extremity hypertension with weak femoral pulses and cool lower legs. Which defect is most likely?

- A. Coarctation of the aorta
- B. Atrial septal defect
- C. Patent ductus arteriosus
- D. Tetralogy of Fallot

A is correct. A narrowed aortic segment creates high pressure proximal to it (arms) and reduced flow distal to it (legs) — the classic arm–leg blood pressure and pulse discrepancy.

Arterial vs venous — the highest-yield comparison in the vascular unit

	Peripheral arterial disease (PAD)	Chronic venous insufficiency
Problem	Blood can't get in — atherosclerotic obstruction	Blood can't get out — incompetent valves, pooling
Pain	Intermittent claudication — cramping with walking, relieved by rest ; severe disease hurts at rest and improves with dangling the leg	Dull ache and heaviness, worse with standing, relieved by elevation
Skin	Cool, pale, shiny, hairless ; thick nails	Warm, brown hemosiderin staining , edema, stasis dermatitis
Pulses	Diminished or absent	Present
Ulcers	Toes, heels, lateral malleolus ; deep, "punched out," pale base, very painful	Medial malleolus ; shallow, irregular, moist, granulating, less painful
Positioning	Dangle the leg (gravity assists arterial inflow)	Elevate the leg (gravity assists venous return)

MNEMONIC Arterial = **Away and up hurts; dangle it down**. Venous = **Down hurts; put it up**. If a question gives you a leg ulcer plus a position that relieves it, that alone identifies the disease.

Other vascular conditions

- **Hypertension** — **primary/essential** (~90–95%, no single cause) vs **secondary** (renal disease, renal artery stenosis, endocrine, medications). Target-organ damage: heart (LVH, failure, CAD), brain (stroke), kidney (nephrosclerosis, CKD), eye (retinopathy), vessels (aneurysm, PAD).
- **Aneurysm** — weakened wall dilates. Abdominal aortic aneurysm may be a **pulsatile abdominal mass**; rupture causes sudden severe back/abdominal pain and shock. **Do not palpate deeply** if suspected.
- **DVT** — unilateral calf swelling, warmth, tenderness, erythema. **The feared consequence is pulmonary embolism** — the Week 9 clot becomes the Week 10 emergency.

- **Raynaud phenomenon** — episodic vasospasm with cold/stress: white (ischemia) → blue (deoxygenation) → red (reperfusion).

Wound healing — phases and what derails them

1. **Hemostasis** — vasoconstriction and clot formation seal the wound.
2. **Inflammation** (~days 1–5) — neutrophils then macrophages clear debris and bacteria; redness, heat, swelling, pain.
3. **Proliferation** (~days 4–21) — fibroblasts lay collagen, angiogenesis, granulation tissue, epithelialization, contraction.
4. **Remodeling / maturation** (weeks–years) — collagen reorganizes; scar gains strength but **never exceeds ~80%** of original.

Factor	Why it delays healing
Diabetes mellitus	Hyperglycemia impairs neutrophil function (infection risk), microvascular disease limits perfusion, and neuropathy means injuries go unnoticed — three deficits at once
Poor perfusion / PAD	Oxygen and nutrients cannot reach the wound; oxygen is required for collagen cross-linking
Infection	Prolongs the inflammatory phase and consumes resources
Corticosteroids	Suppress inflammation and fibroblast/collagen synthesis
Smoking	Nicotine vasoconstricts; carbon monoxide displaces oxygen from hemoglobin
Malnutrition	Protein for collagen, vitamin C for cross-linking, zinc for epithelialization; low albumin also causes edema that impairs diffusion
Advanced age, obesity, pressure	Slower cell turnover; adipose is poorly vascularized; unrelieved pressure blocks flow

Pearl — healing by intention. **Primary** = clean approximated edges (surgical incision), minimal scar. **Secondary** = large open wound heals by granulation and contraction, more scarring, longer. **Tertiary/delayed primary** = left open (often for infection), then closed later.

RED FLAG — WOUND COMPLICATIONS

Dehiscence = separation of wound edges. **Evisceration** = organs protrude through the opening — a surgical emergency; cover with sterile saline-moistened gauze and do not push organs back.

PRACTICE

Q1. A patient's leg pain worsens with standing and improves with elevation; the ankle is edematous with brown discoloration and a shallow ulcer at the medial malleolus. Which condition is this?

- A. Chronic venous insufficiency
- B. Peripheral arterial disease
- C. Acute arterial occlusion
- D. Raynaud phenomenon

A is correct. Relief with elevation, hemosiderin (brown) staining, edema, and a **medial malleolar** ulcer are the venous picture. Arterial disease has the opposite pattern: relief with dangling, pale cool skin, absent pulses, and painful punched-out toe ulcers.

PRACTICE

Q2. Which mechanism best explains delayed wound healing in diabetes mellitus?

- A. Hyperglycemia impairs leukocyte function and microvascular disease reduces oxygen delivery to the wound
- B. Excessive collagen deposition
- C. Overactive inflammatory response shortens healing
- D. Increased capillary density at the wound

A is correct. Diabetes attacks healing on three fronts — impaired immune defense, poor perfusion, and neuropathy that delays detection of the injury in the first place.

PRACTICE

Q3. During which phase of healing is granulation tissue formed and collagen first deposited?

- A. Proliferative phase
- B. Hemostasis
- C. Inflammatory phase
- D. Remodeling phase

A is correct. Fibroblasts build the collagen matrix and new capillaries form granulation tissue during **proliferation**. Remodeling later reorganizes that collagen for strength.

WEEK 10

Pulmonary Function

Getting air in, getting gas across, and keeping pH stable — plus the ABG method that turns a hard question into a checklist.

13 Ventilation, Diffusion & the Drive to Breathe

Process	Meaning	Fails in
Ventilation	Air moving in and out of alveoli	Airway obstruction, chest wall/neuromuscular disease, CNS depression
Diffusion	Gas crossing the alveolar–capillary membrane	Fibrosis, pulmonary edema, emphysema (lost surface area)
Perfusion	Blood flowing past alveoli	Pulmonary embolism, low cardiac output

Key idea — surfactant. Made by **type II alveolar cells**, surfactant **reduces surface tension** so alveoli stay open at end-expiration and require less work to reinflate. Without it, alveoli collapse (atelectasis), compliance falls, and the work of breathing skyrockets — the exact problem in **neonatal RDS** and **ARDS**.

Control of breathing

Receptor	Location	Senses	Role
Central chemoreceptors	Medulla	CO₂ / pH of CSF	The primary, moment-to-moment drive to breathe — rising CO ₂ is the strongest stimulus
Peripheral chemoreceptors	Carotid and aortic bodies	Low O₂ (mainly below PaO ₂ ~60), also pH	Backup drive; becomes dominant in chronic CO ₂ retention

Pearl — CO₂ is the boss. Ask "what normally makes you breathe?" and the answer is **carbon dioxide**, not oxygen. In chronic CO₂ retainers (advanced COPD) the medulla becomes desensitized and the **hypoxic drive** takes over — which is why oxygen in those patients is titrated to a target rather than given wide open.

Compliance and resistance

- **Compliance** = how easily the lung stretches. ↓ in **fibrosis, ARDS, pulmonary edema** (stiff lungs — hard to inflate). ↑ in **emphysema** (destroyed elastic tissue — easy to inflate, hard to *deflate*).

- **Airway resistance** rises steeply as radius falls — halving a radius raises resistance enormously. This is why a small amount of edema or mucus is trivial in an adult airway and **critical in an infant airway**.

PRACTICE

Q1. Which stimulus provides the primary drive to breathe in a healthy adult?

- A. Rising arterial carbon dioxide detected by central chemoreceptors
- B. Falling arterial oxygen detected by carotid bodies
- C. Stretch receptors in the chest wall
- D. Rising bicarbonate in plasma

A is correct. Central chemoreceptors in the medulla respond to CO₂-driven changes in CSF pH. Oxygen becomes the primary driver only in chronic retainers or when PaO₂ falls below about 60 mmHg.

PRACTICE

Q2. Which consequence follows most directly from surfactant deficiency?

- A. Increased alveolar surface tension causing alveolar collapse and increased work of breathing
- B. Increased lung compliance
- C. Bronchial smooth muscle constriction
- D. Excess mucus production

A is correct. Surfactant lowers surface tension. Without it, alveoli collapse at end-expiration, compliance falls, and each breath must overcome collapse — the mechanism in neonatal RDS and ARDS.

PRACTICE

Q3. In emphysema, lung compliance is increased. Why does this still impair ventilation?

- A. Loss of elastic recoil makes exhalation passive and incomplete, trapping air
- B. The lungs become too stiff to inflate
- C. Surfactant production doubles
- D. Airway resistance falls to zero

A is correct. Elastic recoil is what drives normal passive exhalation. Destroy it and air is trapped → hyperinflation, barrel chest, and increased residual volume — easy in, hard out.

14 Acid–Base & ABGs — a 4-Step Method



More $\text{CO}_2 \rightarrow$ more acid $\rightarrow$ **lower pH**. Blow off $\text{CO}_2 \rightarrow$ less acid $\rightarrow$ **higher pH**.

Value	Normal range	Controlled by
pH	7.35 – 7.45	—
PaCO ₂	35 – 45 mmHg	Lungs (fast — minutes)
HCO ₃ [−]	22 – 26 mEq/L	Kidneys (slow — hours to days)
PaO ₂	80 – 100 mmHg	Oxygenation (read separately from acid–base)
SaO ₂	95 – 100%	Oxygenation

THE 4-STEP ABG METHOD

1. **Look at pH.** <7.35 = **acidosis**. >7.45 = **alkalosis**. (7.35–7.45 with abnormal values = **fully compensated** — then use which side of 7.40 it sits on.)
2. **Look at PaCO₂.** If it moves **opposite** to pH, the problem is **respiratory**.
3. **Look at HCO₃[−].** If it moves in the **same** direction as pH, the problem is **metabolic**.
4. **Assess compensation.** Other value normal = **uncompensated**. Other value abnormal but pH still off = **partially compensated**. pH back in range with both abnormal = **fully compensated**.

MNEMONIC ROME — **R**espiratory **O**pposite (pH and CO₂ move in opposite directions), **M**etabolic **E**qual (pH and HCO₃ move in the same direction).

Disorder	pH	PaCO ₂	HCO ₃ ⁻	Common causes	Compensated by
Respiratory acidosis	↓	↑	n → ↑	Hypoventilation: COPD, opioid overdose, neuromuscular weakness, airway obstruction, atelectasis	Kidneys retain HCO ₃ ⁻ (slow)
Respiratory alkalosis	↑	↓	n → ↓	Hyperventilation: anxiety/panic, pain, fever, early sepsis, high altitude, early PE	Kidneys excrete HCO ₃ ⁻
Metabolic acidosis	↓	n → ↓	↓	DKA, lactic acidosis (shock), renal failure, severe diarrhea (bicarbonate loss)	Lungs blow off CO ₂ — Kussmaul respirations (fast)
Metabolic alkalosis	↑	n → ↑	↑	Vomiting / NG suction (acid loss), diuretics, excess antacids, hypokalemia	Lungs retain CO ₂ (hypoventilation)

Pearl — vomiting vs diarrhea. Vomiting loses gastric **acid** → **metabolic alkalosis**. Diarrhea loses intestinal **bicarbonate** → **metabolic acidosis**. Same organ system, opposite disorders, decided entirely by which end the fluid leaves from.

CROSS-WEEK THREAD Acid–base is where Week 10 and Week 11 meet. The **lungs** compensate fast (minutes); the **kidneys** compensate slowly (hours–days) but more completely. A fully compensated respiratory acidosis therefore implies a **chronic** process — the kidneys have had days to work.

PRACTICE

Q1. ABG: pH 7.28, PaCO₂ 58, HCO₃ 25. Which disorder is this?

- A. Uncompensated respiratory acidosis
- B. Metabolic acidosis
- C. Compensated respiratory acidosis
- D. Respiratory alkalosis

A is correct. pH is low (acidosis); CO₂ is high and moves **opposite** to pH → respiratory. Bicarbonate is still normal, so the kidneys have not yet compensated — **uncompensated**, and therefore acute.

PRACTICE

Q2. A patient in diabetic ketoacidosis breathes deeply and rapidly (Kussmaul respirations). Why?

- A. The lungs are blowing off CO₂ to compensate for metabolic acidosis
- B. The patient has respiratory alkalosis as the primary disorder
- C. Ketones directly stimulate the airway
- D. Bicarbonate is being retained by the lungs

A is correct. Ketoacids consume bicarbonate → metabolic acidosis. The medulla drives deep rapid breathing to eliminate CO₂ (carbonic acid), pushing pH back up — respiratory compensation for a metabolic problem.

PRACTICE

Q3. ABG: pH 7.37, PaCO₂ 60, HCO₃ 34 in a patient with advanced COPD. How is this best described?

- A. Fully compensated respiratory acidosis
- B. Uncompensated metabolic alkalosis
- C. Partially compensated metabolic alkalosis
- D. Normal ABG

A is correct. pH is back within range but on the acidic side of 7.40, with both CO₂ and HCO₃ markedly elevated. The kidneys have had time to retain bicarbonate — the signature of a **chronic** respiratory acidosis.

15 Obstructive vs Restrictive Disease

Key idea — one sentence separates them. Obstructive = trouble getting air OUT (airways narrowed; air trapped; volumes go *up*). **Restrictive = trouble getting air IN** (lungs or chest wall stiff; volumes go *down*).

	OBSTRUCTIVE	RESTRICTIVE
Core defect	↑ airway resistance, air trapping	↓ lung compliance / expansion
Total lung capacity & residual volume	↑ Increased (hyperinflation)	↓ Decreased
FEV₁/FVC ratio	↓ Reduced — the hallmark	Normal or increased (both fall together)
Hallmark sign	Prolonged expiration, wheezing, barrel chest	Rapid shallow breathing, fine crackles
Examples	Asthma, chronic bronchitis, emphysema, cystic fibrosis, bronchiectasis	Pulmonary fibrosis, ARDS, pneumoconiosis, obesity, kyphoscoliosis, neuromuscular disease

Asthma

A **reversible** obstructive disease with three simultaneous components:

1. **Bronchoconstriction** — smooth muscle spasm
2. **Airway inflammation and edema** — the underlying chronic process
3. **Mucus hypersecretion** — plugging

RED FLAG — THE SILENT CHEST

A wheezing asthmatic is moving air. If wheezing **disappears** while the patient worsens, becomes drowsy, or the ABG shows a **rising CO₂** (a "normal" CO₂ in a tachypneic asthmatic is ominous), airflow has become so poor that no sound is produced — **impending respiratory failure**.

COPD — two phenotypes

	Chronic bronchitis ("blue bloater")	Emphysema ("pink puffer")
Lesion	Airway problem: mucus gland hypertrophy, chronic inflammation	Alveolar problem: destruction of alveolar walls and elastic tissue
Definition	Productive cough ≥3 months for 2 consecutive years	Permanent enlargement of airspaces distal to terminal bronchioles
Appearance	Cyanotic, edematous (cor pulmonale), overweight, copious sputum	Thin, barrel chest, pursed-lip breathing , tripod position, minimal sputum
Gas exchange	Marked hypoxemia and CO ₂ retention early	Relatively preserved until late — maintained by increased work

Pearl — why pursed-lip breathing works. It creates **back-pressure** that splints collapsing airways open during exhalation, allowing trapped air to escape. Patients discover it on their own because it genuinely improves gas exchange.

CROSS-WEEK THREAD Chronic hypoxemia from COPD → pulmonary vasoconstriction → **pulmonary hypertension** → right ventricular hypertrophy and failure = **cor pulmonale** (Week 9), while the same hypoxemia drives EPO → **secondary polycythemia** (Week 8). One lung disease, three chapters.

PRACTICE

Q1. Which finding indicates an obstructive rather than a restrictive pattern?

- A. Decreased FEV₁/FVC ratio with increased residual volume
- B. Decreased total lung capacity with a normal FEV₁/FVC ratio
- C. Reduced lung compliance
- D. Rapid shallow breathing with fine crackles

A is correct. Obstruction limits **expiratory flow**, so FEV₁ falls disproportionately to FVC (ratio drops) and unexhaled air accumulates (residual volume rises). Restriction lowers all volumes while the ratio stays normal.

PRACTICE

Q2. A severe asthmatic becomes quieter with diminishing wheeze, drowsiness, and a PaCO_2 that has risen from 30 to 44. What does this indicate?

- A. Impending respiratory failure from severe airflow limitation and tiring respiratory muscles
- B. Resolution of the attack
- C. Development of respiratory alkalosis
- D. Successful bronchodilation

A is correct. A "normalizing" CO_2 in a struggling asthmatic means ventilation is failing, not improving. Combined with a quiet chest and altered mental status, this is a respiratory arrest warning.

PRACTICE

Q3. Which mechanism explains barrel chest in emphysema?

- A. Destruction of elastic tissue causes air trapping and chronic hyperinflation
- B. Increased mucus production expands the thorax
- C. Fibrosis contracts the lung
- D. Pleural fluid accumulation

A is correct. Without elastic recoil, air cannot be fully exhaled. Chronic retention of that volume permanently increases the anteroposterior chest diameter.

16 Hypoxemia & V/Q Mismatch

Key idea — V/Q is a ratio of ventilation (V) to perfusion (Q). Ideal exchange needs both. Break either one and oxygenation fails — but in *opposite* ways.

Pattern	What's wrong	Classic cause	Responds to supplemental O ₂ ?
Low V/Q (shunt)	Perfusion without ventilation — blood passes unventilated alveoli	Pneumonia, atelectasis, pulmonary edema, ARDS	Poorly — a true shunt is refractory to oxygen because the blood never contacts air
High V/Q (dead space)	Ventilation without perfusion — air reaches alveoli with no blood flow	Pulmonary embolism , severe emphysema	Partially — but the blocked segment cannot participate at all

MNEMONIC Shunted blood is **Stuck** (perfused, not ventilated — O₂ won't help). **Dead space is Dry** (ventilated, not perfused). Pneumonia = shunt. PE = dead space.

The five mechanisms of hypoxemia

1. **V/Q mismatch** — the most common cause overall.
2. **Shunt** — the extreme of low V/Q; refractory hypoxemia.
3. **Hypoventilation** — low O₂ *with* high CO₂ (opioids, neuromuscular disease).
4. **Diffusion impairment** — thickened membrane (fibrosis, edema); worst with exertion.
5. **Low inspired oxygen** — altitude, enclosed space.

Pearl — hypoxemia vs hypoxia. **Hypoxemia** = low oxygen in the **blood** (a PaO₂/SaO₂ number). **Hypoxia** = inadequate oxygen at the **tissue**. Anemia (Week 8), carbon monoxide poisoning, and shock (Week 9) all cause tissue **hypoxia with a normal PaO₂** — the blood is saturated, the delivery is not.

RED FLAG — PULMONARY EMBOLISM

Sudden dyspnea, pleuritic chest pain, tachycardia, tachypnea, anxiety, sometimes hemoptysis — often in a patient with Virchow's triad risk factors (post-op, immobile, malignancy, estrogen, recent long travel). The **hypoxemia with a low CO₂** (from hyperventilation) is characteristic. A large PE causes **obstructive shock**.

Key idea — late signs of chronic hypoxemia: cyanosis (requires substantial deoxygenated hemoglobin — and therefore appears **late, and may never appear in severe anemia**), **digital clubbing** (chronic), **polycythemia**, **pulmonary hypertension** → **cor pulmonale**.

PRACTICE

Q1. A patient with lobar pneumonia remains hypoxemic despite high-flow oxygen. Which mechanism explains this?

- A. Shunt — blood perfuses consolidated alveoli that cannot be ventilated, so it never contacts oxygen
- B. Dead space ventilation
- C. Hypoventilation
- D. Low inspired oxygen concentration

A is correct. Consolidation fills alveoli with exudate. Blood flowing past them cannot pick up oxygen no matter how much is delivered to *other* alveoli — the defining feature of refractory, shunt-type hypoxemia.

PRACTICE

Q2. A pulmonary embolus creates which V/Q abnormality?

- A. High V/Q (dead space) — alveoli are ventilated but not perfused
- B. Low V/Q (shunt)
- C. Normal V/Q with diffusion impairment
- D. Hypoventilation

A is correct. The clot stops blood flow while air still enters — ventilation without perfusion. That ventilated-but-useless volume is physiologic **dead space**.

PRACTICE

Q3. Why may cyanosis be absent in a severely anemic, hypoxic patient?

- A. Cyanosis requires a substantial absolute amount of deoxygenated hemoglobin, which anemia cannot supply
- B. Anemia increases oxygen saturation
- C. Anemic blood is more alkaline
- D. Anemia dilates cutaneous vessels

A is correct. The blue color depends on the **quantity** of deoxygenated hemoglobin, not the percentage. With little hemoglobin present, the tissue can be profoundly hypoxic while the skin still looks pale rather than blue.

17 Alveolar & Pleural Disorders

Disorder	Mechanism	Key findings
Pneumonia	Infection → inflammatory exudate fills alveoli → consolidation → shunt	Fever, productive cough, pleuritic pain, crackles, dullness to percussion, bronchial breath sounds , ↑ tactile fremitus
Atelectasis	Alveolar collapse — from obstruction (mucus plug), compression, or hypoventilation (post-op, pain, immobility)	↓ breath sounds over the area, hypoxemia; prevented by deep breathing, incentive spirometry, early ambulation
ARDS	Diffuse alveolar–capillary membrane injury → protein-rich fluid floods alveoli + surfactant is inactivated → widespread collapse and shunt	Refractory hypoxemia , bilateral infiltrates ("white-out"), stiff noncompliant lungs, not from cardiac failure. Triggers: sepsis, aspiration, trauma, pancreatitis, near-drowning
Pulmonary edema (cardiogenic)	↑ pulmonary capillary hydrostatic pressure from left heart failure	Crackles, pink frothy sputum , orthopnea, ↑ BNP
Pneumothorax	Air enters the pleural space → loss of negative pressure → lung collapses	Sudden pleuritic pain, dyspnea, absent breath sounds and hyperresonance on the affected side
Pleural effusion	Fluid in the pleural space	Dullness to percussion, ↓ breath sounds, ↓ fremitus. Empyema = pus; hemothorax = blood

RED FLAG — TENSION PNEUMOTHORAX

A one-way valve lets air in but not out. Rising pressure collapses the lung **and shifts the mediastinum**, kinking the great veins. Signs: severe dyspnea, **tracheal deviation AWAY from the affected side**, distended neck veins, absent breath sounds, hypotension. This is **obstructive shock** and requires immediate needle decompression — do not wait for imaging.

Pearl — percussion decides air vs fluid. Both pneumothorax and effusion give absent/decreased breath sounds. **Hyperresonant** = air (pneumothorax). **Dull** = fluid (effusion, and also consolidation). One tap separates them.

Pearl — ARDS vs cardiogenic pulmonary edema. Both flood the alveoli and both give bilateral infiltrates. **Cardiogenic:** high pressure pushing fluid out — expect JVD, S3, elevated BNP, responds to diuresis. **ARDS:** a **leaky membrane** — normal cardiac filling pressures, refractory hypoxemia, and diuresis does not fix it.

PRACTICE

Q1. Which finding distinguishes a tension pneumothorax from a simple pneumothorax?

- A. Tracheal deviation away from the affected side with hypotension and distended neck veins
- B. Absent breath sounds on the affected side
- C. Sudden pleuritic chest pain
- D. Dyspnea and tachypnea

A is correct. The first three occur in both. Only **tension** physiology shifts the mediastinum and obstructs venous return, producing tracheal deviation, JVD, and shock.

PRACTICE

Q2. Which mechanism underlies ARDS?

- A. Diffuse alveolar-capillary membrane injury with protein-rich fluid leak and surfactant inactivation
- B. Elevated left atrial pressure pushing fluid into alveoli
- C. Bronchial smooth muscle constriction
- D. Air entering the pleural space

A is correct. ARDS is a **permeability** problem, not a pressure problem. The leak plus surfactant loss produces widespread collapse, stiff lungs, and hypoxemia that oxygen alone cannot correct.

PRACTICE

Q3. A postoperative patient who is splinting from incisional pain develops decreased breath sounds at the lung bases and hypoxemia. Which complication is most likely?

- A. Atelectasis from hypoventilation and retained secretions
- B. Pneumothorax
- C. Pulmonary embolism
- D. Pleural effusion

A is correct. Shallow splinted breathing prevents full alveolar expansion and clearance of secretions → collapse. This is why incentive spirometry, coughing, and early ambulation are standard postoperative care.

Key idea — why children decompensate faster. Narrower airways (small amounts of edema cause large resistance increases), **more compliant chest wall** (retractions instead of effective volume), **fewer alveoli** and less reserve, **higher metabolic rate** (oxygen consumed faster), **obligate nose-breathing** in young infants, and **faster fatigue** of respiratory muscles.

Condition	Mechanism	Signature
Neonatal RDS	Surfactant deficiency from prematurity → alveolar collapse	Preterm infant with grunting, nasal flaring, retractions, tachypnea in the first hours; treated with surfactant and antenatal steroids
Transient tachypnea of the newborn	Retained fetal lung fluid	Term/near-term, often cesarean birth ; tachypnea resolving in 24–72 h
Meconium aspiration	Meconium obstructs airways, inactivates surfactant, causes chemical pneumonitis	Post-term or stressed infant, meconium-stained fluid, respiratory distress
Bronchiolitis (RSV)	Inflammation and mucus plugging of bronchioles	Infant <2 yr: wheezing, retractions, copious secretions, apnea in young infants ; supportive care
Croup (laryngotracheobronchitis)	Viral (parainfluenza) subglottic edema	Barking/seal-like cough, inspiratory stridor, hoarseness ; gradual onset, often at night; improves with cool humidified air
Epiglottitis	Bacterial (classically <i>H. influenzae</i> type b) supraglottic infection	Rapid onset high fever, drooling, dysphagia, distress, tripod position , muffled voice — and no cough
Cystic fibrosis	Autosomal recessive CFTR chloride-channel defect → thick, viscous secretions	Recurrent pulmonary infections, obstruction, pancreatic insufficiency → steatorrhea and failure to thrive , salty sweat, meconium ileus

RED FLAG — EPIGLOTTITIS

Do not inspect the throat, use a tongue blade, or upset the child — laryngospasm can cause complete airway obstruction. Keep the child calm and upright with a parent and get airway expertise immediately. The mnemonic is the **4 D's: Drooling, Dysphagia, Dysphonia, Distress**.

Pearl — croup vs epiglottitis in one line. Croup barks; epiglottitis drools. Croup is viral, gradual, with a loud barking cough. Epiglottitis is bacterial, abrupt, toxic-appearing, with drooling and *no* cough.

Pearl — retractions are the pediatric warning sign. A compliant chest wall means a struggling child pulls the soft tissue inward (substernal, intercostal, supraclavicular) instead of moving air. **Nasal flaring, grunting, and head bobbing** are the same message: work of breathing is unsustainable.

PRACTICE

Q1. A 3-year-old has abrupt high fever, drooling, muffled voice, and sits leaning forward; there is no cough. What is the priority?

- A. Keep the child calm and upright and obtain immediate airway support without examining the throat
- B. Examine the pharynx with a tongue depressor
- C. Place the child supine for assessment
- D. Give cool humidified air and reassess in an hour

A is correct. This is **epiglottitis**. Any throat manipulation or agitation risks laryngospasm and total obstruction. Airway management takes priority over examination.

PRACTICE

Q2. Which mechanism causes neonatal respiratory distress syndrome in a premature infant?

- A. Insufficient surfactant production leading to alveolar collapse and increased work of breathing
- B. Retained fetal lung fluid
- C. Aspiration of meconium
- D. Viral bronchiolar inflammation

A is correct. Type II cells do not produce adequate surfactant until later in gestation. Without it, surface tension collapses alveoli with every breath — hence grunting, an infant's attempt to generate its own back-pressure.

PRACTICE

Q3. Why does a small amount of airway edema cause far more obstruction in an infant than in an adult?

- A. The infant's smaller airway radius means the same edema causes a disproportionately large rise in resistance
- B. Infants have stiffer chest walls
- C. Infants have more alveoli
- D. Infants have a slower metabolic rate

A is correct. Resistance rises steeply as radius falls, so 1 mm of edema narrows an infant airway proportionally far more than an adult's — the physiologic basis for croup and bronchiolitis being pediatric emergencies.

WEEK 11

Renal & Urinary Function

Filtration, the hormones the kidney controls, and why kidney failure damages six other systems.

Four processes, in order

Process	Where	What happens
Filtration	Glomerulus → Bowman capsule	Pressure-driven; water and small solutes pass. Proteins and cells should NOT — their presence in urine means the filter is damaged
Reabsorption	Mostly proximal tubule; loop; distal	Reclaims glucose, amino acids, most Na^+ and water, bicarbonate
Secretion	Tubules	Adds K^+ , H^+ , drugs, and wastes into the filtrate
Excretion	Collecting duct → urine	Final output = filtered – reabsorbed + secreted

Key idea — GFR is the number that defines kidney function. Normal is roughly **90–125 mL/min**. **Creatinine** is the better marker (produced steadily by muscle, filtered and not reabsorbed) — a **rising creatinine means a falling GFR**. BUN also rises but is less specific: it also increases with dehydration, high protein intake, and GI bleeding.

Autoregulation — the kidney protects its own filtration

- **Myogenic response** — afferent arteriole constricts when pressure rises, dilates when it falls.
- **Tubuloglomerular feedback** — the macula densa senses filtrate sodium and adjusts arteriolar tone.
- **RAAS** — juxtaglomerular cells release **renin** when renal perfusion, sodium delivery, or blood pressure falls.

Pearl — autoregulation has a floor. It maintains GFR over a wide pressure range, but below roughly a mean arterial pressure of 60–70 mmHg it fails and GFR collapses. That is the moment **prerenal AKI** begins — and why sustained hypotension in shock injures kidneys first.

Renal hormones — and the disease each one causes when lost

Hormone	Action	Lost in CKD → consequence
Renin → angiotensin II → aldosterone	Vasoconstriction; Na ⁺ and water retention, K⁺ excretion	Dysregulated → hypertension and fluid overload
Erythropoietin	Stimulates red cell production	↓ → normocytic anemia
Activated vitamin D (calcitriol)	Kidney performs the final activation step → gut absorbs calcium	↓ → hypocalcemia → secondary hyperparathyroidism → renal osteodystrophy
ADH (from pituitary, acts on collecting duct)	Water reabsorption — concentrates urine	Impaired concentrating ability early in CKD → nocturia
ANP / BNP (from heart, act on kidney)	Promote sodium and water excretion	Overwhelmed → edema persists

CROSS-WEEK THREAD The kidney is the hub of this exam. It supplies EPO (Week 8 anemia), it drives RAAS (Week 9 heart failure), it compensates acid–base with the lungs (Week 10), and it activates vitamin D for bone (Week 14). Learn the four hormones and four other chapters get easier.

PRACTICE

Q1. Which finding most directly indicates a decline in glomerular filtration rate?

- A. Rising serum creatinine
- B. Rising urine specific gravity
- C. Falling serum potassium
- D. Rising hemoglobin

A is correct. Creatinine is produced at a steady rate and cleared almost entirely by filtration, so serum creatinine and GFR move inversely. BUN also rises but is affected by hydration, diet, and GI bleeding.

PRACTICE

Q2. Why does chronic kidney disease cause hypocalcemia?

- A. The kidney can no longer activate vitamin D, so intestinal calcium absorption falls
- B. The kidney excretes excess calcium
- C. Parathyroid hormone production ceases
- D. Calcium is consumed by red cell production

A is correct. Calcitriol requires a final activation step in the kidney. Without it the gut cannot absorb calcium → hypocalcemia → PTH rises (secondary hyperparathyroidism) → calcium is pulled from bone → renal osteodystrophy.

PRACTICE

Q3. Protein and red blood cells appear in the urine. Which structure is damaged?

- A. The glomerular filtration membrane
- B. The collecting duct
- C. The renal pelvis
- D. The urethra

A is correct. The glomerulus normally excludes cells and large proteins by size and charge. Their appearance in urine means the filtration barrier itself has been injured.

Kidney stones (urolithiasis)

- **Calcium oxalate** stones are the **most common**. Others: uric acid (gout, high purine), struvite (associated with urea-splitting infections), cystine (genetic).
- **The universal risk factor is dehydration** — concentrated urine lets crystals precipitate. Also immobility, high sodium/oxalate intake, hyperparathyroidism, family history.
- **Presentation:** sudden severe **colicky flank pain radiating to the groin, hematuria**, nausea/vomiting, restlessness (the patient cannot find a comfortable position — unlike peritonitis, where patients lie still).

Key idea — obstruction backs up. A stone blocking outflow raises pressure behind it → **hydronephrosis** (dilation of the renal pelvis) → compression of nephrons → **postrenal AKI**. Obstruction plus infection is urgent: stagnant urine behind a blockage rapidly becomes urosepsis.

Urinary tract infection

	Cystitis (lower UTI)	Pyelonephritis (upper UTI)
Location	Bladder	Kidney / renal pelvis
Symptoms	Dysuria, frequency, urgency, suprapubic discomfort , cloudy or foul urine	All of the above plus fever, chills, flank pain, costovertebral angle tenderness, nausea/vomiting
Urine finding	Bacteriuria, pyuria, nitrites, leukocyte esterase	WBC casts — casts form in tubules, so they prove kidney involvement

Pearl — the discriminators. Casts localize to the kidney (WBC casts = pyelonephritis, RBC casts = glomerulonephritis, muddy brown casts = acute tubular necrosis). **Systemic signs** (fever, chills, CVA tenderness) also mean upper tract. Simple cystitis is typically afebrile.

Key idea — risk factors follow anatomy and flow. *E. coli* from the perineum is the usual organism. Women are at higher risk because of a **shorter urethra** closer to the anus. Anything that causes **stasis or instrumentation** raises risk: catheters, BPH-related retention (Week 12), pregnancy, stones, neurogenic bladder, diabetes.

RED FLAG — THE OLDER ADULT

In older adults a UTI may present as **new confusion, falls, or functional decline** with little or no fever and no classic dysuria. Do not require the textbook triad before considering infection.

PRACTICE

Q1. A patient has fever, chills, flank pain, costovertebral angle tenderness, and WBC casts in the urine. Which condition is this?

- A. Pyelonephritis — infection involving the kidney
- B. Simple cystitis
- C. Urethritis
- D. Nephrolithiasis

A is correct. Systemic signs plus CVA tenderness indicate upper tract disease, and **WBC casts form in renal tubules** — they cannot originate in the bladder. That single finding localizes the infection.

PRACTICE

Q2. Which factor most contributes to kidney stone formation?

- A. Dehydration producing concentrated urine that allows crystals to precipitate
- B. Excess water intake
- C. Low urinary calcium
- D. Frequent voiding

A is correct. Supersaturation is the common pathway for every stone type, which is why aggressive hydration is both the primary treatment and the primary prevention.

PRACTICE

Q3. A ureteral stone causes progressive dilation of the renal pelvis. What is this called, and what does it threaten?

- A. Hydronephrosis — back-pressure compresses nephrons and can cause postrenal acute kidney injury
- B. Glomerulonephritis — immune injury to the glomerulus
- C. Nephrotic syndrome — massive proteinuria
- D. Acute tubular necrosis from nephrotoxins

A is correct. Obstructed outflow transmits pressure backward into the collecting system. Sustained hydronephrosis compresses and destroys nephrons, so relieving obstruction is time-sensitive.

21 Acute Kidney Injury — Pre/Intra/Postrenal

Key idea — locate the problem relative to the kidney. **Prerenal** = the problem is *before* the kidney (not enough blood arriving). **Intrarenal** = the problem is *in* the kidney tissue. **Postrenal** = the problem is *after* the kidney (urine cannot get out). Prerenal and postrenal are reversible if caught early; intrarenal means tissue damage.

	PRERENAL	INTRARENAL	POSTRENAL
Problem	↓ renal perfusion	Direct nephron damage	Obstruction to outflow
Causes	Hypovolemia, hemorrhage, dehydration, heart failure , shock, sepsis, renal artery stenosis	Acute tubular necrosis from ischemia or nephrotoxins (aminoglycosides, contrast dye, NSAIDs, myoglobin from rhabdomyolysis); glomerulonephritis; acute interstitial nephritis	BPH , stones, tumor, strictures, blocked catheter, neurogenic bladder
BUN : creatinine	> 20:1 (urea is reabsorbed with the avidly retained sodium and water)	~10–15:1 (both rise together)	Variable
Urine sodium	↓ Low — the kidney is working hard to conserve sodium	↑ High — damaged tubules cannot conserve sodium	Variable
Urine sediment	Bland; concentrated urine, high specific gravity	Muddy brown granular casts (ATN); RBC casts (glomerulonephritis)	Bland
Fix	Restore perfusion/volume	Remove the toxin, support — takes time to regenerate	Relieve the obstruction

MNEMONIC Prerenal = **thirsty kidney** (holding onto sodium and water → low urine Na, high BUN:Cr, concentrated urine). Intrarenal = **broken kidney** (can't hold anything → high urine Na, dilute urine, casts).

The four phases of AKI

1. **Onset/initiation** — the insult occurs; hours to days.
2. **Oliguric** — urine output falls (<400 mL/day). **The dangerous phase:** fluid overload, **hyperkalemia**, metabolic acidosis, rising BUN/creatinine.
3. **Diuretic** — tubules recover filtration before concentrating ability, so urine pours out. **The risk flips** to dehydration, **hypokalemia**, and hyponatremia.
4. **Recovery** — gradual return of function over weeks to months; may be incomplete.

RED FLAG — HYPERKALEMIA

The most immediately life-threatening consequence of AKI. Potassium is normally excreted by the kidney; when excretion stops, it accumulates and destabilizes cardiac conduction → peaked T waves, widened QRS, and **cardiac arrest**. In the oliguric phase, potassium is what kills first.

CROSS-WEEK THREAD Nearly every prior week can cause AKI: heart failure and shock (Week 9, prerenal), rhabdomyolysis from crush injury or compartment syndrome (Week 14, intrarenal via myoglobin), acute hemolytic transfusion reaction and DIC (Week 8), and BPH (Week 12, postrenal). When a question gives you kidney failure, ask *which upstream chapter caused it*.

PRACTICE

Q1. A dehydrated patient has oliguria, BUN:creatinine of 28:1, low urine sodium, and high urine specific gravity. Which category of AKI is this?

- A. Prerenal — reduced perfusion with intact tubular function conserving sodium and water
- B. Intrarenal from acute tubular necrosis
- C. Postrenal from obstruction
- D. Chronic kidney disease

A is correct. The high BUN:Cr ratio, low urine sodium, and concentrated urine all show the tubules are **working correctly** to conserve volume — the kidney is starved, not damaged. Restoring perfusion reverses it.

PRACTICE

Q2. Which finding suggests acute tubular necrosis rather than prerenal azotemia?

- A. Muddy brown granular casts with high urine sodium and dilute urine
- B. Low urine sodium with high specific gravity
- C. BUN:creatinine ratio above 20:1
- D. Rapid response to a fluid bolus

A is correct. Damaged tubules shed cellular debris (muddy brown casts) and lose the ability to reabsorb sodium or concentrate urine — the opposite profile from prerenal disease.

PRACTICE

Q3. During the oliguric phase of AKI, which electrolyte abnormality poses the most immediate threat to life?

- A. Hyperkalemia causing cardiac dysrhythmias
- B. Hypernatremia
- C. Hypocalcemia
- D. Hypomagnesemia

A is correct. With excretion halted, potassium accumulates rapidly and destabilizes cardiac membrane potential — peaked T waves, QRS widening, then arrest. It is the priority monitoring parameter in oliguric AKI.

22 Glomerular Disease — Nephritic vs Nephrotic

Key idea — one filter, two ways to break it. **Nephritic** = the glomerulus is **inflamed**, so **cells** leak through (blood in urine). **Nephrotic** = the glomerulus is **leaky to protein**, so **protein** pours out. Inflammation vs permeability.

	NEPHRITIC syndrome	NEPHROTIC syndrome
Core lesion	Glomerular inflammation	Glomerular permeability to protein
Urine	Hematuria — tea/cola-colored, RBC casts ; proteinuria mild (<3.5 g/day)	Massive proteinuria >3.5 g/day , frothy urine; hematuria minimal
Blood	↑ BUN/creatinine, hypertension	Hypoalbuminemia, hyperlipidemia
Edema	Mild, often periorbital, from salt/water retention	Severe, generalized — low oncotic pressure (periorbital, ascites, anasarca)
Classic example	Post-streptococcal glomerulonephritis — 1–3 weeks after strep pharyngitis or impetigo; immune complexes deposit in the glomerulus	Minimal change disease (most common in children); membranous nephropathy; diabetic nephropathy
Added risks	Fluid overload, hypertensive complications	Infection (immunoglobulins lost in urine) and thrombosis (anticoagulant proteins lost)

MNEMONIC Nephritic = Inflammation and blood (RBC casts). Nephrotic = prOteinuria and edema. If the stem emphasizes cola-colored urine and hypertension, it's nephritic; if it emphasizes frothy urine, swelling, and low albumin, it's nephrotic.

Pearl — post-streptococcal glomerulonephritis is an immune-complex disease, not an infection of the kidney. The strep infection is already resolving; antigen–antibody complexes deposit in the glomerulus and trigger inflammation. That is why there is a **1–3 week lag** and why antibiotics at that point do not reverse the kidney injury.

CROSS-WEEK THREAD Nephrotic syndrome is a Week 11 disease that produces a Week 9 sign (edema, from low oncotic pressure) and a Week 8 risk (thrombosis, from urinary loss of

anticoagulant proteins). Same low-albumin logic as cirrhosis in Week 13.

PRACTICE

Q1. A child develops cola-colored urine, periorbital edema, and hypertension two weeks after a sore throat. Which condition is this?

- A. Post-streptococcal glomerulonephritis — immune complex deposition causing glomerular inflammation
- B. Nephrotic syndrome from minimal change disease
- C. Acute pyelonephritis
- D. Nephrolithiasis

A is correct. The lag after strep infection, hematuria, hypertension, and mild edema define the **nephritic** picture. Minimal change disease would show heavy proteinuria and severe generalized edema without hematuria.

PRACTICE

Q2. Which mechanism causes the severe edema of nephrotic syndrome?

- A. Urinary loss of albumin lowers plasma oncotic pressure so fluid remains in the interstitium
- B. Increased capillary permeability throughout the body
- C. Lymphatic obstruction
- D. Right-sided heart failure

A is correct. Massive proteinuria depletes albumin, the main determinant of oncotic pull. Without it, capillary filtrate is not drawn back into the vessel — the same mechanism as edema in cirrhosis.

PRACTICE

Q3. Why are patients with nephrotic syndrome at increased risk for infection and thrombosis?

- A. Immunoglobulins and anticoagulant proteins are lost in the urine along with albumin
- B. The kidney overproduces clotting factors
- C. Hyperlipidemia suppresses the immune system
- D. Edema mechanically blocks lymph nodes

A is correct. The damaged filter is not selective for albumin alone. Loss of immunoglobulins impairs defense; loss of antithrombin and related regulators tips the balance toward clotting.

23 Chronic Kidney Disease & Its Six Consequences

CKD is **progressive, irreversible** nephron loss over months to years, staged by GFR (stage 5 / end-stage <15 mL/min requires dialysis or transplant). The leading causes are **diabetes mellitus** and **hypertension**.

Key idea — CKD symptoms are just "the kidney's jobs, undone." Rather than memorizing a symptom list, list what the kidney does and remove each function. Every consequence below is the failure of one specific job.

#	Kidney job lost	Consequence	Clinical picture
1	Excrete water and sodium	Fluid overload	Edema, hypertension, heart failure, pulmonary edema
2	Excrete potassium	Hyperkalemia	Dysrhythmias, muscle weakness — a leading cause of death
3	Excrete acid / regenerate bicarbonate	Metabolic acidosis	Kussmaul respirations, fatigue, bone demineralization
4	Produce erythropoietin	Normocytic anemia	Fatigue, pallor, dyspnea (Week 8)
5	Activate vitamin D + excrete phosphate	Renal osteodystrophy	↓Ca → ↑PTH → bone resorption → fractures, bone pain (Week 14)
6	Excrete nitrogenous waste	Uremia	Nausea, anorexia, metallic taste, pruritus, uremic frost , confusion, asterixis, pericarditis , platelet dysfunction → bleeding

Pearl — the calcium/phosphate seesaw. Failing kidneys **retain phosphate** and **cannot activate vitamin D**. High phosphate binds calcium and low calcitriol reduces absorption → **hypocalcemia** → parathyroid glands ramp up (**secondary hyperparathyroidism**) → PTH pulls calcium out of bone → weak bone. Phosphate up, calcium down, PTH up — that trio is the CKD bone signature.

Pearl — why CKD patients bleed. Uremic toxins impair **platelet function** even when the platelet **count** is normal — a Week 8 concept (function vs number) explaining a Week 11 finding.

RED FLAG — AKI VS CKD

Both raise BUN and creatinine. **AKI is abrupt and potentially reversible; CKD is gradual and irreversible.** Clues to chronicity: **anemia, small shrunken kidneys** on imaging, **bone disease**, and long-standing diabetes or hypertension — none of which develop overnight.

PRACTICE

Q1. Which explains the anemia of chronic kidney disease?

- A. Reduced erythropoietin production by damaged kidneys
- B. Iron malabsorption in the stomach
- C. Autoimmune destruction of red cells
- D. Vitamin B₁₂ deficiency

A is correct. With nephron mass lost, EPO production falls and the marrow is never stimulated. The resulting anemia is **normocytic normochromic** — the cells produced are normal, there are simply too few.

PRACTICE

Q2. Which laboratory trio is characteristic of chronic kidney disease-related bone disease?

- A. High phosphate, low calcium, high parathyroid hormone
- B. Low phosphate, high calcium, low parathyroid hormone
- C. Normal phosphate, normal calcium, high vitamin D
- D. High calcium, high vitamin D, low parathyroid hormone

A is correct. Retained phosphate plus failure to activate vitamin D drives calcium down; the parathyroids respond by raising PTH, which resorbs bone — renal osteodystrophy.

PRACTICE

Q3. Which finding best suggests chronic rather than acute kidney disease?

- A. Anemia and small, shrunken kidneys on imaging
- B. Elevated serum creatinine
- C. Oliguria
- D. Hyperkalemia

A is correct. Creatinine, oliguria, and hyperkalemia occur in both. Anemia and kidney atrophy require **months to years** to develop, so they indicate a chronic process.

Condition	Mechanism	Signature
Vesicoureteral reflux	Incompetent ureterovesical junction lets urine flow backward toward the kidney	Recurrent UTIs; risk of reflux nephropathy and scarring — why recurrent pediatric UTIs get investigated
Posterior urethral valves	Obstructing membrane in the male urethra	Weak stream, distended bladder, hydronephrosis in a male infant
Polycystic kidney disease	Genetic; fluid-filled cysts replace functional tissue	Bilateral flank masses, hypertension, progressive kidney failure
Minimal change disease	Podocyte injury → protein leak	Most common nephrotic syndrome in children ; periorbital edema, frothy urine, responds well to steroids
Post-streptococcal GN	Immune complexes after strep infection	Nephritic picture 1–3 weeks after pharyngitis/impetigo; usually self-limited in children
Hemolytic uremic syndrome	Shiga toxin (<i>E. coli</i> O157:H7, undercooked beef) damages endothelium → microthrombi shred red cells and consume platelets	Triad: hemolytic anemia + thrombocytopenia + acute kidney injury , following bloody diarrhea
Wilms tumor (nephroblastoma)	Embryonal kidney malignancy, typically ages 2–5	Firm, unilateral, non-tender abdominal mass , often found incidentally — DO NOT PALPATE the abdomen (risk of seeding tumor cells)

RED FLAG — DO NOT PALPATE

Suspected **Wilms tumor**: post a sign and avoid repeated abdominal palpation. This is the pediatric equivalent of "do not deeply palpate a suspected abdominal aortic aneurysm" (Week 9) — the exam question is testing whether you know that *assessment itself* can cause harm.

Key idea — why infants dehydrate so fast. A higher **percentage of body weight is water** and more of it is **extracellular** (easily lost), the **metabolic rate and turnover** are higher, **immature kidneys cannot concentrate urine** efficiently, the body surface area to mass ratio is larger, and infants **cannot ask for fluid**. Vomiting and diarrhea that would inconvenience an adult can be life-threatening in an infant.

Pearl — HUS ties three weeks together. One toxin produces a Week 8 problem (hemolytic anemia + thrombocytopenia), a Week 11 problem (AKI), and follows a Week 13 problem (infectious bloody diarrhea). Recognize the triad after bloody diarrhea and the diagnosis is made.

PRACTICE

Q1. A toddler develops hemolytic anemia, thrombocytopenia, and acute kidney injury several days after bloody diarrhea. Which condition is this?

- A. Hemolytic uremic syndrome
- B. Post-streptococcal glomerulonephritis
- C. Minimal change disease
- D. Wilms tumor

A is correct. Shiga toxin from *E. coli* O157:H7 injures endothelium; microthrombi shear red cells and consume platelets while occluding glomerular capillaries. The triad after bloody diarrhea is diagnostic.

PRACTICE

Q2. A firm, non-tender unilateral abdominal mass is found in a 3-year-old. Which action is contraindicated?

- A. Repeated deep abdominal palpation, because it may disseminate tumor cells
- B. Measuring blood pressure
- C. Obtaining a urinalysis
- D. Documenting the finding

A is correct. This presentation suggests **Wilms tumor**. Palpation risks rupturing the capsule and seeding malignant cells, so abdominal palpation is avoided once the mass is suspected.

PRACTICE

Q3. Why does vesicoureteral reflux lead to kidney damage?

- A. Urine flows backward toward the kidney, carrying bacteria upward and causing recurrent infection and scarring
- B. It obstructs the urethra completely
- C. It destroys erythropoietin-producing cells directly
- D. It causes massive proteinuria

A is correct. Retrograde flow delivers bladder bacteria to the renal pelvis, producing repeated pyelonephritis. Each episode scars nephrons, and cumulative scarring causes hypertension and chronic kidney disease.

WEEK 12

Reproductive Systems

Hormonal cycles and what happens when they misfire — plus the structural disorders and the two urologic emergencies.

25 The Menstrual Cycle & PCOS

The axis

Hypothalamus (**GnRH**) → anterior pituitary (**FSH** and **LH**) → ovary (**estrogen** and **progesterone**) → negative feedback back to hypothalamus and pituitary.

Phase	Days (28-day model)	Dominant hormone	What happens
Menstrual	1–5	All low	Endometrium is shed
Proliferative (follicular)	6–13	Estrogen (from the developing follicle)	Endometrium rebuilds and thickens ; FSH matures the follicle
Ovulation	~14	LH surge	Follicle ruptures and releases the oocyte; may cause mittelschmerz and a small temperature rise
Secretory (luteal)	15–28	Progesterone (from the corpus luteum)	Endometrium becomes secretory and receptive to implantation

Key idea — why menstruation happens. If fertilization does not occur, the **corpus luteum degenerates** → **progesterone falls** → the endometrium loses its hormonal support → spiral arteries constrict → the lining is shed. **Menstruation is a progesterone-withdrawal event.**

MNEMONIC Estrogen builds it up; progesterone holds it steady; withdrawal lets it go.

"**Progesterone** = **pro**-gestation" — it maintains the lining for pregnancy.

Menstrual and ovulatory disorders

Term	Meaning
Primary amenorrhea	Menses never began by the expected age
Secondary amenorrhea	Previously regular cycles stop (pregnancy first, then thyroid, prolactin, PCOS, weight loss, athletic training, stress)
Dysmenorrhea	Painful menses — prostaglandin -mediated uterine contraction and ischemia (why NSAIDs, prostaglandin inhibitors, help)
Menorrhagia	Heavy or prolonged bleeding — consider fibroids, adenomyosis, bleeding disorders (Week 8 vWD)
PMS / PMDD	Luteal-phase symptoms; PMDD is the severe form with marked mood disturbance

PCOS — polycystic ovary syndrome

Defined by a combination of **hyperandrogenism**, **oligo- or anovulation**, and **polycystic-appearing ovaries**.

Insulin resistance → hyperinsulinemia → ovarian **androgen** production rises and sex hormone-binding globulin falls → more free androgen → **follicles arrest** before ovulating → **anovulation** + multiple small follicles + **hirsutism, acne**.

Feature	Explanation
Irregular or absent menses, infertility	No ovulation → no corpus luteum → no progesterone
Hirsutism, acne, male-pattern hair thinning	Excess androgen
Obesity, acanthosis nigricans	Insulin resistance
↑ LH:FSH ratio	Altered pituitary pulsation
Long-term risks: type 2 diabetes, metabolic syndrome, cardiovascular disease, endometrial hyperplasia and carcinoma	Chronic unopposed estrogen without progesterone means the endometrium proliferates but is never shed on schedule

Pearl — the unopposed-estrogen concept is worth points. Anything that gives estrogen stimulation *without* progesterone opposition (PCOS/anovulation, obesity, estrogen-only therapy, early menarche/late menopause, nulliparity) raises **endometrial cancer** risk. Progesterone is the brake.

PRACTICE

Q1. Which hormonal event directly triggers ovulation?

- A. A surge in luteinizing hormone
- B. A fall in estrogen
- C. A rise in progesterone from the corpus luteum
- D. A fall in follicle-stimulating hormone

A is correct. Rising estrogen from the mature follicle briefly switches to **positive** feedback and triggers the **LH surge**, which ruptures the follicle. Progesterone rises *after* ovulation, from the corpus luteum.

PRACTICE

Q2. Why does PCOS increase the risk of endometrial hyperplasia?

- A. Chronic anovulation means estrogen stimulates the endometrium without the opposing effect of progesterone
- B. Excess progesterone thickens the endometrium
- C. Elevated FSH causes endometrial growth
- D. Androgens directly stimulate endometrial cells

A is correct. Without ovulation there is no corpus luteum and therefore no progesterone. The endometrium proliferates continuously under unopposed estrogen and is never regularly shed — hyperplasia and eventually carcinoma risk.

PRACTICE

Q3. Which underlying metabolic abnormality drives the hormonal changes in PCOS?

- A. Insulin resistance with compensatory hyperinsulinemia increasing ovarian androgen production
- B. Primary thyroid failure
- C. Adrenal insufficiency
- D. Growth hormone excess

A is correct. Hyperinsulinemia stimulates ovarian androgen output and lowers sex hormone-binding globulin, raising free androgen. This is why weight management and insulin-sensitizing therapy improve both the metabolic and reproductive features.

Disorder	Mechanism	Signature
Endometriosis	Endometrial tissue implants outside the uterus; it still responds to hormones, so it bleeds cyclically into tissue → inflammation, adhesions, scarring	Cyclic pelvic pain worst around menses, dyspareunia , dyschezia, infertility . Severity of pain does not correlate with extent of disease
Leiomyoma (fibroid)	Benign smooth-muscle tumor of the myometrium; estrogen-dependent — grows in reproductive years, shrinks after menopause	Menorrhagia , pelvic pressure/bulk symptoms, urinary frequency, enlarged irregular uterus; may impair fertility
Adenomyosis	Endometrial tissue grows into the myometrium	Heavy painful menses with a uniformly enlarged, tender, boggy uterus
Ovarian cysts	Follicular or corpus luteum cysts	Usually asymptomatic; sudden severe unilateral pain suggests rupture or ovarian torsion
Pelvic inflammatory disease	Ascending infection (commonly chlamydia, gonorrhea) from cervix to uterus, tubes, ovaries	Lower abdominal pain, cervical motion tenderness , fever, discharge. Scarring → infertility and ectopic pregnancy risk
Cervical cancer	Persistent high-risk HPV (types 16 and 18) → dysplasia → carcinoma	Often asymptomatic early — detected by screening; later postcoital bleeding. Vaccine-preventable
Breast cancer	Risk with age, family history, BRCA1/BRCA2 , prolonged estrogen exposure	Hard, fixed, painless, irregular mass ; skin dimpling, nipple retraction, peau d'orange — contrast with the smooth, mobile, often tender mass of benign disease

Pearl — benign vs malignant masses in one line. Benign tends to be **smooth, mobile, well-defined, often tender**. Malignant tends to be **hard, fixed, irregular, painless**, with skin or nipple changes. This distinction is a direct callback to the oncology principles from earlier in the course.

CROSS-WEEK THREAD PID scars the tubes → a fertilized ovum cannot travel → **ectopic pregnancy**. A ruptured ectopic causes rapid intra-abdominal hemorrhage → **hypovolemic shock** (Week 9). Sudden unilateral pelvic pain with a missed period and hemodynamic instability is a surgical emergency.

PRACTICE

Q1. Which mechanism explains cyclic pelvic pain and infertility in endometriosis?

- A. Ectopic endometrial tissue responds to cyclic hormones, bleeding into tissue and causing inflammation, adhesions, and scarring
- B. The uterus enlarges from benign smooth muscle tumors
- C. The endometrium fails to respond to estrogen
- D. Ascending bacterial infection scars the fallopian tubes

A is correct. The misplaced tissue is hormonally responsive but has nowhere to drain. Repeated cyclic bleeding causes chronic inflammation and adhesions that distort pelvic anatomy — hence both the pain and the infertility.

PRACTICE

Q2. Why do uterine fibroids typically shrink after menopause?

- A. They are estrogen-dependent and estrogen levels fall sharply
- B. They become malignant and are removed
- C. Progesterone rises after menopause
- D. Uterine blood flow increases

A is correct. Leiomyomas depend on ovarian estrogen for growth. Once ovarian production ceases, the stimulus is withdrawn and the tumors regress.

PRACTICE

Q3. Which long-term consequence of pelvic inflammatory disease results from tubal scarring?

- A. Infertility and increased risk of ectopic pregnancy
- B. Endometrial hyperplasia
- C. Ovarian androgen excess
- D. Uterine leiomyoma formation

A is correct. Inflammation scars the fallopian tubes. A blocked tube prevents fertilization; a partially blocked tube may trap a fertilized ovum, causing implantation in the tube — an ectopic pregnancy.

27 Infections & Menopause

Infection	Cause	Discharge / clue
Vulvovaginal candidiasis	<i>Candida</i> overgrowth — after antibiotics , in diabetes, pregnancy, immunosuppression	Thick, white, "cottage cheese"; intense itching; normal pH
Bacterial vaginosis	Loss of protective lactobacilli → overgrowth of anaerobes	Thin gray-white, fishy odor , elevated pH; itching usually mild
Trichomoniasis	Protozoan, sexually transmitted	Frothy yellow-green , malodorous, irritation
Chlamydia / gonorrhea	Bacterial STIs	Often asymptomatic in women — which is exactly why they progress to PID and cause silent tubal damage

Pearl — why antibiotics cause yeast infections. Normal **lactobacilli** keep vaginal pH acidic and suppress *Candida*. Broad-spectrum antibiotics kill the lactobacilli, the pH rises, and yeast overgrows — a direct example of normal flora as a defense mechanism.

Menopause

Permanent cessation of menses (confirmed after **12 consecutive months**) as ovarian follicles are depleted.

Follicles depleted → **estrogen and progesterone fall** → loss of negative feedback → **FSH and LH rise** (an elevated FSH is the lab signature).

Consequence	Mechanism
Hot flashes, night sweats	Estrogen withdrawal destabilizes hypothalamic thermoregulation
Vaginal dryness, atrophy, dyspareunia	Loss of estrogen-dependent epithelium and lubrication; also ↑ UTI risk
Accelerated bone loss → osteoporosis	Estrogen normally restrains osteoclasts ; without it resorption outpaces formation (Week 14)
Rising cardiovascular risk	Loss of estrogen's favorable effects on lipids and endothelium
Sleep disturbance, mood changes, urinary symptoms	Multifactorial estrogen withdrawal

RED FLAG — POSTMENOPAUSAL BLEEDING

Any vaginal bleeding after menopause is **abnormal until proven otherwise** and must be evaluated for **endometrial carcinoma**. This is a frequently tested "which finding requires follow-up" answer.

PRACTICE

Q1. Which hormonal pattern characterizes menopause?

- A. Low estrogen with elevated FSH and LH
- B. High estrogen with low FSH
- C. High progesterone with low LH
- D. Low FSH with low LH

A is correct. Depleted follicles cannot produce estrogen. With negative feedback removed, the pituitary drives FSH and LH upward — an elevated FSH is the classic confirmatory finding.

PRACTICE

Q2. Why does osteoporosis risk rise sharply after menopause?

- A. Estrogen normally restrains osteoclast activity; its loss allows bone resorption to exceed formation
- B. Calcium intake automatically decreases
- C. Vitamin D activation stops
- D. Osteoblasts increase excessively

A is correct. Estrogen is a brake on bone resorption. Removing the brake accelerates bone loss, most rapidly in the first years after menopause — the link between Week 12 and Week 14.

PRACTICE

Q3. A woman develops thick white discharge and intense itching after a course of antibiotics. Which mechanism explains this?

- A. Antibiotics eliminated protective lactobacilli, allowing *Candida* overgrowth
- B. Antibiotics directly stimulate *Candida* growth
- C. The antibiotics raised estrogen levels
- D. A sexually transmitted protozoan infection developed

A is correct. Normal flora suppresses fungal overgrowth by maintaining an acidic environment and competing for resources. Eliminating that flora removes the restraint on *Candida*.

Physiology essentials

- **Leydig cells** produce **testosterone**; **Sertoli cells** support **spermatogenesis**. FSH drives Sertoli function; LH drives Leydig function.
- The testes sit outside the body because **spermatogenesis requires a temperature below core** — the reason undescended testes and varicoceles impair fertility.
- **Erection is parasympathetic; ejaculation is sympathetic**. Nitric oxide → cGMP → smooth muscle relaxation → arterial inflow and venous trapping.

MNEMONIC "Point and Shoot": Parasympathetic = **P**oint (erection). Sympathetic = **S**hoot (emission/ejaculation).

Key idea — aging male changes: gradual decline in testosterone, reduced sperm quality, **prostate enlargement (BPH)**, longer time to achieve erection and longer refractory period. These are gradual and physiologic — distinguish them from pathology like sudden erectile dysfunction (often vascular) or obstructive urinary retention.

Testicular and scrotal disorders

Disorder	Mechanism	Signature
Testicular torsion	Spermatic cord twists → arterial supply cut off → ischemia and infarction	SURGICAL EMERGENCY. Sudden severe unilateral pain, nausea, high-riding testis , absent cremasteric reflex , pain not relieved by elevation. Salvage window is roughly 6 hours
Epididymitis	Infection/inflammation of the epididymis	Gradual onset, fever, pain relieved by scrotal elevation (Prehn sign), cremasteric reflex present
Cryptorchidism	Failure of testicular descent	Empty scrotum; raises risk of infertility and testicular cancer even after surgical correction
Varicocele	Dilated pampiniform venous plexus, usually left side (drainage anatomy)	"Bag of worms" feel, dull ache, warmer scrotum → infertility
Hydrocele	Serous fluid collection around the testis	Painless swelling that transilluminates
Testicular cancer	Germ cell tumor; peak ages 15–35	Painless, firm, unilateral mass that does not transilluminate; highly curable if caught early — the case for self-examination

RED FLAG — TORSION VS EPIDIDYMITIS

Both cause acute scrotal pain. **Torsion:** sudden, severe, absent cremasteric reflex, elevation does **not** help → go to surgery. **Epididymitis:** gradual, febrile, cremasteric reflex intact, elevation **relieves** pain. Getting this wrong costs a testis, which is exactly why exams ask it.

PRACTICE

Q1. An adolescent has sudden severe scrotal pain, a high-riding testis, and an absent cremasteric reflex. What is the priority?

- A. Immediate surgical evaluation — testicular torsion causes ischemia and testicular loss within hours
- B. Oral antibiotics and scrotal elevation
- C. Warm compresses and reassessment tomorrow
- D. Transillumination and routine referral

A is correct. Torsion strangles the testicular blood supply. The salvage window is roughly six hours, so this is a time-critical surgical emergency — not an infection to be treated medically.

PRACTICE

Q2. Which finding suggests testicular cancer rather than a hydrocele?

- A. A firm painless mass that does not transilluminate
- B. A soft swelling that transilluminates
- C. Pain relieved by elevation
- D. Fever and dysuria

A is correct. Fluid transmits light; solid tumor does not. A firm, painless, non-transilluminating mass in a man aged 15–35 must be evaluated for malignancy.

PRACTICE

Q3. Why does a varicocele impair fertility?

- A. Pooled venous blood raises scrotal temperature, and spermatogenesis requires a temperature below core body temperature
- B. It obstructs the vas deferens mechanically
- C. It reduces testosterone production to zero
- D. It causes autoimmune destruction of sperm

A is correct. The scrotum's external position exists to keep the testes cool. Dilated veins deliver warm blood and defeat that cooling, impairing sperm production — the same principle as cryptorchidism.

29 BPH vs Prostate Cancer & Erectile Dysfunction

	Benign prostatic hyperplasia	Prostate cancer
Zone involved	Central / transitional zone — wraps the urethra	Peripheral zone — away from the urethra
Why symptoms differ	Compresses the urethra early → obstructive symptoms while still benign	Grows away from the urethra → often asymptomatic until advanced
Exam	Smooth, symmetrically enlarged, rubbery	Hard, irregular nodule , asymmetric
Symptoms	Hesitancy, weak stream, straining, incomplete emptying, nocturia, frequency, urgency, dribbling, retention	Late: similar urinary symptoms, hematuria, and bone pain from metastasis (spine/pelvis)
Marker	PSA may be mildly elevated	PSA typically more elevated (not specific alone)

Key idea — anatomy explains everything here. BPH is benign but symptomatic *early* because it grows **inward against the urethra**. Prostate cancer is dangerous but silent *early* because it grows **outward in the peripheral zone**. "More symptoms" does not mean "more serious."

CROSS-WEEK THREAD BPH → incomplete emptying → **urinary stasis** → recurrent **UTI** and **stone** formation, and progressive obstruction → back-pressure → **hydronephrosis** → **postrenal AKI** (Week 11). A Week 12 diagnosis is a common Week 11 cause.

Pearl — prostate cancer metastasizes to bone. New back or pelvic pain in an older man with urinary symptoms is a classic stem for bone metastasis — and a Week 14 crossover (pathologic fracture risk).

Erectile dysfunction

Category	Mechanism	Examples
Vascular (most common)	Impaired arterial inflow — the same atherosclerosis as CAD/PAD	Hypertension, hyperlipidemia, smoking, diabetes. ED can be an early warning of systemic vascular disease
Neurogenic	Damaged parasympathetic pathways	Diabetic autonomic neuropathy , spinal cord injury, pelvic/prostate surgery, multiple sclerosis
Hormonal	Low testosterone, hyperprolactinemia, thyroid disease	Reduced libido accompanies the ED
Medication-induced	Drug effects on vascular tone or neurotransmission	Antihypertensives, antidepressants, alcohol
Psychogenic	Anxiety, depression, relationship stress	Nocturnal and morning erections preserved — the discriminator from organic causes

Pearl — diabetes hits erectile function twice: microvascular disease impairs inflow **and** autonomic neuropathy impairs the parasympathetic signal. Two mechanisms, one disease — the same double hit seen in diabetic wound healing (Week 9).

PRACTICE

Q1. Why does benign prostatic hyperplasia cause obstructive urinary symptoms while early prostate cancer often does not?

- A. BPH arises in the central zone surrounding the urethra, whereas cancer typically arises in the peripheral zone
- B. BPH grows much faster than cancer
- C. Cancer never affects the prostate size
- D. BPH invades the bladder wall directly

A is correct. Location, not aggressiveness, determines early symptoms. Central growth compresses the urethra immediately; peripheral growth can enlarge substantially before it obstructs anything.

PRACTICE

Q2. Untreated BPH with chronic incomplete bladder emptying most directly increases the risk of which complication?

- A. Urinary tract infection, stones, and hydronephrosis with postrenal kidney injury
- B. Nephrotic syndrome
- C. Glomerulonephritis
- D. Renal artery stenosis

A is correct. Stagnant residual urine promotes bacterial growth and crystal precipitation, and sustained back-pressure dilates the collecting system, damaging nephrons.

PRACTICE

Q3. Which finding suggests a psychogenic rather than organic cause of erectile dysfunction?

- A. Preserved nocturnal and early-morning erections
- B. Absent morning erections with gradual onset
- C. Coexisting peripheral arterial disease
- D. Long-standing poorly controlled diabetes

A is correct. Intact spontaneous erections during sleep prove the vascular and neurologic machinery works. Organic causes impair erections in all settings and typically begin gradually.

WEEK 13

Digestive Function

Move it, break it down, absorb it — and what happens when the liver, gallbladder, or pancreas quits.

Motility

- **Peristalsis** propels; **segmentation** mixes.
- Sphincters gate flow: **lower esophageal** (reflux barrier), **pyloric** (gastric emptying), **ileocecal** (prevents backflow), **anal** (continence).
- Motility is largely **parasympathetic-driven** ("rest and digest"); sympathetic activation slows it — why stress, pain, opioids, and shock cause ileus.

Secretion — who makes what

Source	Product	Function — and what deficiency causes
Parietal cells (stomach)	HCl and intrinsic factor	Acid activates pepsinogen and kills microbes; intrinsic factor is required to absorb B₁₂ → loss causes pernicious anemia (Week 8)
Chief cells	Pepsinogen → pepsin	Protein digestion
Mucous cells	Mucus and bicarbonate	The mucosal barrier — when it fails, acid injures the stomach itself → ulcer
Pancreas (exocrine)	Amylase, lipase, proteases + bicarbonate	The main digestive enzymes; bicarbonate neutralizes gastric acid. Loss → steatorrhea and fat-soluble vitamin (A, D, E, K) deficiency
Liver → gallbladder	Bile	Emulsifies fat so lipase can work. Obstruction → fat malabsorption + clay-colored stools

Key idea — absorption is a small-intestine job. Villi and microvilli create enormous surface area. Two locations are worth memorizing: **iron** is absorbed in the **duodenum**, and **vitamin B₁₂** (bound to intrinsic factor) in the **terminal ileum** — which is why Crohn disease of the terminal ileum causes B₁₂ deficiency.

Malabsorption

Cause	Mechanism	Clue
Celiac disease	Immune reaction to gluten → villous atrophy → lost surface area	Diarrhea, steatorrhea, weight loss, iron/folate deficiency, dermatitis herpetiformis; resolves on a gluten-free diet
Lactose intolerance	Lactase deficiency → undigested lactose draws water in and is fermented	Osmotic diarrhea , bloating, cramping, flatus after dairy
Pancreatic insufficiency	No lipase → fat not digested	Steatorrhea — bulky, greasy, foul, floating stools; fat-soluble vitamin deficiency
Crohn disease / bowel resection	Inflamed or absent absorptive surface	Site-specific deficiencies (terminal ileum → B ₁₂ , bile salts)

Pearl — steatorrhea points to fat digestion, not fat intake. Greasy floating stools mean either **no lipase** (pancreatic insufficiency) or **no bile** (obstruction) — and both cause deficiency of the fat-soluble vitamins **A, D, E, K**. Low vitamin K → **prolonged PT/INR** (Week 8); low vitamin D → **osteomalacia** (Week 14).

PRACTICE

Q1. Loss of gastric parietal cells produces which deficiency and why?

- A. Vitamin B₁₂ deficiency, because parietal cells make intrinsic factor required for its absorption
- B. Iron deficiency, because parietal cells absorb iron
- C. Folate deficiency, because parietal cells store folate
- D. Vitamin C deficiency

A is correct. Parietal cells secrete both HCl and **intrinsic factor**. Without intrinsic factor, B₁₂ cannot be absorbed in the terminal ileum → **pernicious anemia**, a macrocytic anemia with neurologic signs.

PRACTICE

Q2. Which mechanism causes diarrhea in lactose intolerance?

- A. Undigested lactose osmotically draws water into the bowel and is fermented by bacteria
- B. Immune-mediated destruction of intestinal villi
- C. Absence of pancreatic lipase
- D. Bacterial invasion of the mucosa

A is correct. Lactase deficiency leaves lactose in the lumen, where it exerts osmotic pull and is fermented to gas and acids — producing the classic osmotic diarrhea with bloating and flatus.

PRACTICE

Q3. A patient with chronic pancreatic insufficiency has bulky, greasy, foul-smelling stools. Which additional problem should be anticipated?

- A. Deficiency of fat-soluble vitamins A, D, E, and K
- B. Excess vitamin K with reduced bleeding risk
- C. Isolated water-soluble vitamin excess
- D. Increased calcium absorption

A is correct. Fat-soluble vitamins require fat digestion and absorption. Without lipase they are lost in stool — vitamin K deficiency prolongs the PT/INR and vitamin D deficiency impairs bone mineralization.

GERD

Incompetent or transiently relaxed **lower esophageal sphincter** lets gastric contents reflux into the esophagus, which has **no protective mucus barrier**.

- **Symptoms:** heartburn (worse lying flat or after large meals), regurgitation, chronic cough, hoarseness, sore throat, dental erosion.
- **Aggravators:** obesity, pregnancy, hiatal hernia, large/fatty meals, alcohol, caffeine, chocolate, nicotine, recumbency after eating.
- **Complications:** esophagitis, stricture, and **Barrett esophagus**.

Key idea — Barrett esophagus is metaplasia. Chronic acid exposure makes squamous esophageal epithelium convert to **columnar (intestinal-type)** epithelium — an **adaptive** change that is more acid-resistant but **premalignant**, raising the risk of **esophageal adenocarcinoma**. Metaplasia → dysplasia → carcinoma is the sequence.

Peptic ulcer disease

An ulcer forms when **aggressive factors overwhelm mucosal defense**.

Aggressive	Protective
<i>H. pylori</i> (the leading cause), NSAIDs (inhibit prostaglandins that maintain mucus and blood flow), acid, pepsin, alcohol, smoking, severe physiologic stress	Mucus and bicarbonate layer, prostaglandins, adequate mucosal blood flow, rapid epithelial renewal

	Gastric ulcer	Duodenal ulcer
Pain timing	Worsens WITH eating (food stimulates acid onto the damaged mucosa)	Relieved BY eating ; returns 2–3 hours after meals and at night
Weight	Often loss (patients avoid eating)	Often stable or gain (patients eat for relief)
Primary defect	Impaired mucosal defense	Excess acid

MNEMONIC Gastric = Greater with food. Duodenal = Down with food (pain goes down when you eat, comes back later).

RED FLAG — ULCER COMPLICATIONS

Hemorrhage (hematemesis, melena, falling hemoglobin, shock). **Perforation** — sudden severe pain with a **rigid, board-like abdomen** and rebound tenderness → peritonitis, a surgical emergency. **Obstruction** from scarring at the pylorus → vomiting, early satiety.

Localizing a GI bleed

Finding	Meaning
Hematemesis (bright red)	Active upper GI bleeding — esophageal varices, ulcer, gastritis
"Coffee-ground" emesis	Upper bleed, slower — blood partially digested by acid
Melena (black, tarry, foul)	Upper bleed — blood has been digested during transit
Hematochezia (bright red per rectum)	Lower GI bleed — or a very brisk upper bleed

Pearl — melena means time. Black tarry stool indicates blood has been in the tract long enough to be digested, which localizes the source to the **upper** tract. Bright red blood has not had that transit time.

CROSS-WEEK THREAD A GI bleed is also a Week 8 and Week 9 problem: acute loss → **normocytic** anemia and possible **hypovolemic shock**; chronic occult loss → **iron-deficiency (microcytic)** anemia. Unexplained iron deficiency in an adult prompts a GI evaluation.

PRACTICE

Q1. A patient reports epigastric pain that is relieved by eating and returns about two hours later and at night. Which ulcer is most likely?

- A. Duodenal ulcer
- B. Gastric ulcer
- C. Esophageal ulcer
- D. Stress ulcer of the fundus

A is correct. Food buffers acid and relieves duodenal pain temporarily; when the stomach empties acidic chyme into the duodenum a few hours later, pain returns. Gastric ulcer pain characteristically **worsens** with eating.

PRACTICE

Q2. Which mechanism explains why NSAIDs cause peptic ulcers?

- A. They inhibit prostaglandins that maintain the protective mucus layer and mucosal blood flow
- B. They directly increase gastrin secretion
- C. They eradicate protective gastric flora
- D. They relax the lower esophageal sphincter

A is correct. Prostaglandins maintain mucosal defense. Blocking them removes protection, so normal acid becomes damaging — a defense failure rather than an acid excess.

PRACTICE

Q3. Chronic GERD leads to replacement of esophageal squamous epithelium with columnar epithelium. What is this called, and why does it matter?

- A. Barrett esophagus — a metaplastic change that is premalignant for adenocarcinoma
- B. Hyperplasia — a benign increase in cell number with no cancer risk
- C. Atrophy — a reduction in cell size that heals spontaneously
- D. Dysplasia — a reversible change requiring no follow-up

A is correct. Metaplasia substitutes a more acid-resistant cell type, but the new tissue is abnormal for that location and carries progression risk to dysplasia and adenocarcinoma — hence surveillance.

32 Inflammatory Bowel Disease & Obstruction

	CROHN DISEASE	ULCERATIVE COLITIS
Location	Anywhere mouth to anus ; most often terminal ileum	Colon and rectum only ; begins at the rectum and extends proximally
Pattern	Skip lesions — patchy, with normal bowel between	Continuous , uninterrupted
Depth	Transmural (full thickness)	Mucosa and submucosa only
Appearance	Cobblestoning , deep linear ulcers, "string sign" of narrowing	Continuous friable mucosa, pseudopolyps
Stool	Diarrhea, often without gross blood ; steatorrhea possible	Bloody diarrhea with mucus , tenesmus
Complications	Fistulas, abscesses, strictures/obstruction , malabsorption, B₁₂ deficiency (terminal ileum)	Toxic megacolon , hemorrhage, markedly ↑ colorectal cancer risk

MNEMONIC Crohn = "**C**obblestones, **C**reeping, **C**omplete-thickness, **s**kip lesions" (and it **C**an be anywhere). **UC** = "**U**lcers in the **C**olon, **C**ontinuous, **b**loody" (and it starts at the rectum). Transmural depth is why **only Crohn makes fistulas**.

Pearl — IBD vs IBS. IBD is **inflammatory and structural**: weight loss, fever, bloody stool, elevated inflammatory markers, night-time symptoms, visible lesions. **IBS** is **functional**: pain relieved by defecation, altered stool form/frequency, no bleeding, no weight loss, normal findings. Alarm features point away from IBS.

Other inflammatory conditions

Condition	Clue
Appendicitis	Periumbilical pain migrating to the right lower quadrant (McBurney point) , anorexia, nausea, rebound tenderness, low-grade fever. Sudden relief of pain suggests perforation — not improvement
Diverticulitis	Left lower quadrant pain, fever, altered bowel habits — inflamed colonic outpouchings
Peritonitis	Rigid board-like abdomen , rebound tenderness, absent bowel sounds; patient lies very still (movement worsens it — unlike renal colic)

Bowel obstruction

	Mechanical	Functional (paralytic ileus)
Cause	Adhesions (#1 small bowel), hernia, tumor (#1 large bowel), intussusception, volvulus, impaction	Loss of peristalsis: post-abdominal surgery , opioids, peritonitis, electrolyte imbalance (hypokalemia), sepsis
Bowel sounds	High-pitched, hyperactive above the block early, then absent as the bowel fatigues	Absent or hypoactive from the start

Key idea — the four cardinal signs of obstruction: abdominal pain (colicky if mechanical), **distension**, **vomiting** (early and bilious in high/small bowel; late and feculent in low/large bowel), and **obstipation** (no stool or flatus). Fluid and gas accumulate proximally and cannot pass.

RED FLAG — THIRD-SPACING AND STRANGULATION

Obstructed bowel sequesters large volumes of fluid into the lumen and wall → **hypovolemia and shock** despite no visible fluid loss, plus electrolyte derangement from vomiting (**metabolic alkalosis**, hypokalemia). If distension compresses the blood supply, the bowel becomes **strangulated** → ischemia, necrosis, perforation, peritonitis. Fever, tachycardia, worsening constant pain, and rising lactate signal that transition.

PRACTICE

Q1. Which feature is characteristic of Crohn disease but not ulcerative colitis?

- A. Transmural inflammation with skip lesions and fistula formation
- B. Continuous involvement beginning at the rectum
- C. Inflammation limited to mucosa and submucosa
- D. Bloody diarrhea with tenesmus

A is correct. Crohn inflammation penetrates the entire wall, which is what allows tracts to burrow into adjacent structures (**fistulas**) and why strictures form. UC is superficial and continuous.

PRACTICE

Q2. Why does mechanical bowel obstruction cause hypovolemia?

- A. Fluid is sequestered ("third-spaced") into the bowel lumen and wall and lost through vomiting
- B. The kidneys excrete excessive sodium
- C. Capillary permeability decreases
- D. Blood is lost into the stool

A is correct. Large volumes accumulate proximal to the obstruction, effectively leaving the circulation. Combined with vomiting, this produces hypovolemia and electrolyte disturbance even though no fluid has visibly left the body.

PRACTICE

Q3. A patient with appendicitis suddenly reports that the severe pain has resolved. What does this most likely indicate?

- A. Perforation of the appendix, which relieves pressure but causes peritonitis
- B. Spontaneous resolution of the inflammation
- C. Successful analgesia
- D. Migration of the pain to the left side

A is correct. Rupture decompresses the distended appendix, briefly relieving pain — then contents spill into the peritoneum and pain returns diffusely with rigidity, fever, and rebound. Sudden relief is a danger sign, not improvement.

Jaundice — three locations, three patterns

	PREhepatic	INTRAhepatic	POSThepatic (obstructive)
Problem	Too much bilirubin produced	Liver cells cannot process it	Bilirubin cannot drain out
Cause	Hemolysis (Week 8), large hematoma resorption	Hepatitis, cirrhosis, drug injury	Gallstone , tumor, stricture in the bile duct
Bilirubin type	Unconjugated (indirect)	Mixed	Conjugated (direct)
Urine / stool	Urine normal; stool normal/dark	Variable	Dark urine + clay-colored (pale) stool + pruritus

Key idea — clay-colored stool localizes the problem instantly. Stool is brown because bile pigment reaches the intestine. **Pale/clay stool means bile is not getting there** → obstruction. The conjugated bilirubin then backs up into blood and spills into urine (dark urine) and deposits in skin (itching).

Cirrhosis — one lesion, two failure modes

Chronic injury (alcohol, hepatitis B/C, nonalcoholic fatty liver disease) → inflammation → **fibrosis and regenerative nodules** → the architecture is destroyed. Two consequences follow, and every sign belongs to one of them.

Failure mode	Mechanism	Signs
1. Portal hypertension Scarred liver obstructs portal blood flow → pressure backs up	Collateral veins engorge	Esophageal varices (life-threatening hemorrhage), caput medusae , hemorrhoids
	↑ hydrostatic pressure + ↓ albumin	Ascites
	Congestion of the spleen	Splenomegaly → thrombocytopenia (platelets sequestered — Week 8)
	Fluid + bacteria	Spontaneous bacterial peritonitis
2. Hepatocellular failure Not enough working liver cells	↓ Albumin synthesis	Edema and ascites (low oncotic pressure)
	↓ Clotting factor synthesis	Prolonged PT/INR , bruising, bleeding
	↓ Detoxification — ammonia accumulates	Hepatic encephalopathy : confusion, sleep reversal, asterixis , coma
	↓ Bilirubin processing; ↓ hormone metabolism	Jaundice , pruritus, spider angiomas, palmar erythema, gynecomastia

RED FLAG — THE BLEEDING TRAP IN CIRRHOSIS

The cirrhotic patient has **varices** (a high-pressure bleeding source), **no clotting factors** (↑INR), and **too few platelets** (splenic sequestration) simultaneously. A variceal bleed is therefore catastrophic — hematemesis in a cirrhotic patient is an immediate emergency.

Pearl — asterixis and ammonia. The liver normally converts ammonia (from protein/bacterial metabolism) to urea. In failure, ammonia crosses into the brain → encephalopathy. **Asterixis** ("flapping tremor" on wrist extension) is the classic sign, and a **GI bleed worsens encephalopathy** because digested blood is a large protein load.

Hepatitis	Transmission	Chronic?
A	Fecal–oral (contaminated food/water)	No — acute only; vaccine available
B	Blood, sexual, perinatal	Can become chronic → cirrhosis, hepatocellular carcinoma; vaccine available
C	Blood (injection drug use, transfusions before screening)	Most likely to become chronic ; no vaccine

PRACTICE

Q1. A patient has jaundice, dark urine, clay-colored stools, and pruritus. Where is the problem?

- A. Posthepatic — obstruction of bile flow, so conjugated bilirubin cannot reach the intestine
- B. Prehepatic — excessive hemolysis
- C. Intrahepatic — hepatocyte injury alone
- D. Renal — impaired bilirubin excretion by the kidney

A is correct. Pale stool proves bile is not reaching the gut. The conjugated (water-soluble) bilirubin backs into blood, is filtered into urine (dark), and deposits in skin causing itching.

PRACTICE

Q2. Which mechanism explains ascites in cirrhosis?

- A. Portal hypertension raises hydrostatic pressure while reduced albumin synthesis lowers oncotic pressure
- B. Increased albumin production pulls fluid into the abdomen
- C. Lymphatic obstruction by gallstones
- D. Excess bile salt secretion

A is correct. Both Starling forces are deranged in the same direction — more push out (portal pressure), less pull in (low albumin). This is the same edema logic as nephrotic syndrome and heart failure, combined.

PRACTICE

Q3. A cirrhotic patient becomes confused with a flapping tremor on wrist extension. Which substance accumulation is responsible?

- A. Ammonia, which the failing liver can no longer convert to urea
- B. Conjugated bilirubin
- C. Albumin
- D. Bicarbonate

A is correct. Ammonia from protein and bacterial metabolism is normally detoxified to urea. In hepatic failure it accumulates and reaches the brain — producing hepatic encephalopathy with asterixis.

Gallstones and cholecystitis

- **Cholelithiasis** = stones in the gallbladder, usually **cholesterol** stones. Classic risk profile: the **4 F's** — **F**emale, **F**orty, **F**ertile, **F**at (also rapid weight loss, family history).
- **Biliary colic**: RUQ pain, often **after a fatty meal** (fat triggers cholecystokinin → gallbladder contracts against the stone), may radiate to the **right shoulder or scapula**.
- **Cholecystitis**: obstruction → inflammation → constant RUQ pain, fever, leukocytosis, **positive Murphy sign** (inspiratory arrest on RUQ palpation).
- **Choledocholithiasis** (stone in the common bile duct) → obstructive jaundice; if infected → **cholangitis (Charcot triad**: RUQ pain, fever/chills, jaundice).

Pancreatitis

Gallstones (obstruct the duct) or **alcohol** (direct toxic injury) → digestive enzymes are activated **inside the pancreas** → **autodigestion** → inflammation, hemorrhage, and fat necrosis.

Finding	Explanation
Severe epigastric pain radiating straight to the back , relieved by leaning forward	Retroperitoneal location of the pancreas
↑ Serum lipase (more specific) and amylase	Enzymes leak from the injured gland
Nausea, vomiting, fever, tachycardia, hypotension	Inflammation + massive third-spacing into the retroperitoneum
Hypocalcemia	Calcium is consumed in fat necrosis (saponification) — a low calcium here is a marker of severity
Cullen sign (periumbilical bruising) / Grey Turner sign (flank bruising)	Retroperitoneal hemorrhage tracking to the skin — severe disease

RED FLAG — SYSTEMIC COMPLICATIONS

Severe pancreatitis is not confined to the abdomen: released inflammatory mediators can cause **ARDS** (Week 10), **hypovolemic shock** from third-spacing (Week 9), **AKI** (Week 11), and **DIC** (Week 8). It is one of the classic ARDS triggers.

Pearl — chronic pancreatitis is a different problem. Repeated injury destroys the gland permanently → **exocrine failure** (steatorrhea, weight loss, fat-soluble vitamin deficiency) and eventually **endocrine failure** (diabetes from islet cell loss). Acute = autodigestion. Chronic = loss of function.

PRACTICE

Q1. Which mechanism underlies acute pancreatitis?

- A. Premature activation of digestive enzymes within the pancreas causing autodigestion
- B. Bacterial invasion of the pancreatic duct
- C. Insufficient insulin secretion
- D. Obstruction of the portal vein

A is correct. Enzymes meant to digest food in the duodenum are activated inside the gland, digesting pancreatic tissue itself — producing inflammation, necrosis, and hemorrhage. Gallstones and alcohol are the two leading triggers.

PRACTICE

Q2. Why does RUQ pain often follow a fatty meal in cholelithiasis?

- A. Fat stimulates cholecystokinin release, causing the gallbladder to contract against an obstructing stone
- B. Fat directly irritates the gallbladder lining
- C. Fat increases gastric acid production
- D. Fat dissolves the gallstones, causing pain

A is correct. CCK signals the gallbladder to eject bile for fat digestion. Contracting against a stone that blocks the outlet produces the classic postprandial colicky pain.

PRACTICE

Q3. Why does hypocalcemia occur in severe acute pancreatitis?

- A. Calcium binds to free fatty acids released during fat necrosis (saponification)
- B. The pancreas normally secretes calcium into blood
- C. The kidneys excrete excess calcium in response to pain
- D. Parathyroid hormone secretion increases

A is correct. Lipase digests peripancreatic fat, releasing fatty acids that chelate calcium into insoluble soaps. A falling calcium level therefore signals extensive necrosis and more severe disease.

Condition	Age	Mechanism & signature
Pyloric stenosis	2–8 weeks	Hypertrophied pyloric muscle obstructs gastric outlet → projectile NON-bilious vomiting (obstruction is <i>above</i> the bile duct), hungry immediately after vomiting, palpable " olive ", visible peristalsis. Loss of gastric acid → hypochloremic metabolic alkalosis with dehydration
Intussusception	3 months–3 years	One segment telescopes into another → obstruction and ischemia. Sudden intermittent severe pain with drawing up of the legs, "currant jelly" stool (blood and mucus), sausage-shaped mass
Hirschsprung disease	Newborn	Absent ganglion cells in a distal colonic segment → no peristalsis there → functional obstruction. Failure to pass meconium within 48 hours , abdominal distension, ribbon-like stools
Necrotizing enterocolitis	Premature infants	Ischemic/inflammatory bowel injury → feeding intolerance, distension, bloody stools , systemic instability — a neonatal emergency
Biliary atresia	Neonate	Bile ducts are absent or obliterated → persistent jaundice beyond 2 weeks with clay-colored stools and dark urine — requires urgent surgical evaluation
Cleft lip / palate	Congenital	Failure of facial process fusion → feeding difficulty (poor suction seal), aspiration risk, otitis media
Infant GER	Infancy	Immature lower esophageal sphincter → effortless spitting up in a thriving infant; usually resolves. Poor weight gain or respiratory symptoms make it pathologic

Key idea — bilious vs non-bilious vomiting localizes the obstruction. Bile enters the duodenum *below* the pylorus. **Non-bilious** vomiting means the block is at or above the pylorus (pyloric stenosis). **Bilious (green)** vomiting means the block is **below** the duodenal papilla — and in a newborn it is an emergency (consider **malrotation with volvulus**).

Pearl — why pyloric stenosis causes alkalosis. Repeated vomiting of gastric contents loses H^+ and Cl^- → **hypochloremic, hypokalemic metabolic alkalosis**. This is the pediatric version of the Week 10 rule: vomiting loses acid → alkalosis; diarrhea loses bicarbonate → acidosis.

PRACTICE

Q1. A 5-week-old has projectile non-bilious vomiting, remains hungry, and has a palpable olive-shaped mass. Which acid-base disturbance is expected?

- A. Hypochloremic metabolic alkalosis from loss of gastric acid
- B. Metabolic acidosis from bicarbonate loss
- C. Respiratory acidosis
- D. Respiratory alkalosis

A is correct. Pyloric stenosis causes repeated vomiting of hydrochloric acid before it reaches the intestine. Losing H^+ and Cl^- produces a hypochloremic (and often hypokalemic) metabolic alkalosis with dehydration.

PRACTICE

Q2. A 9-month-old has sudden episodic severe abdominal pain with leg drawing, a sausage-shaped mass, and stool containing blood and mucus. Which condition is this?

- A. Intussusception — telescoping of one bowel segment into another
- B. Pyloric stenosis
- C. Hirschsprung disease
- D. Appendicitis

A is correct. Telescoping obstructs the lumen and compromises blood supply; sloughed mucosa and blood produce the classic **currant jelly** stool. The intermittent pain pattern reflects peristaltic waves against the obstruction.

PRACTICE

Q3. A newborn fails to pass meconium in the first 48 hours and has progressive abdominal distension. Which mechanism explains this?

- A. Absence of ganglion cells in a distal colon segment prevents peristalsis through that segment
- B. Hypertrophy of the pyloric sphincter
- C. Telescoping of the ileum into the colon
- D. Obliteration of the extrahepatic bile ducts

A is correct. In Hirschsprung disease the aganglionic segment cannot relax or propagate peristalsis, creating a **functional** obstruction. Stool accumulates proximally and the newborn cannot pass meconium.

WEEK 14

Musculoskeletal & Integumentary

Bone, joint, muscle, and skin — the arthritis trio, the fracture emergencies, burns, and the childhood rashes.

36 Bone Disorders, Fractures & Emergencies

Bone is living tissue in constant turnover: **osteoclasts** resorb, **osteoblasts** build. The major disorders differ in **what** is wrong with the bone.

Disorder	Defect	Key clue
Osteoporosis	Too little bone — resorption exceeds formation; matrix and mineral both reduced	Postmenopausal/elderly; fragility fractures (hip, wrist, vertebral compression → height loss, kyphosis); serum calcium normal
Osteomalacia (rickets in children)	Poorly mineralized bone — usually vitamin D deficiency	Soft, bendable bone; bone pain; bowed legs in children
Osteogenesis imperfecta	Genetic type I collagen defect → brittle scaffold	Fractures from birth with minimal force, blue sclerae
Paget disease	Disorganized, excessive remodeling	Enlarged, deformed, structurally weak bone; ↑ alkaline phosphatase
Osteomyelitis	Bone infection (often <i>S. aureus</i>)	Localized pain, fever, warmth; hard to eradicate because bone is poorly penetrated by antibiotics

Key idea — the three bone defects in one line. **Osteoporosis** = **not enough bone** (what's there is normal; calcium labs normal). **Osteomalacia** = **bad mineralization** (a vitamin D problem — soft, not scarce). **Osteogenesis imperfecta** = **bad collagen** (the protein scaffold itself is defective).

CROSS-WEEK THREAD Three earlier chapters converge on bone: **menopause** removes estrogen's brake on osteoclasts (Week 12), **CKD** blocks vitamin D activation and retains phosphate (Week 11), and **fat malabsorption** depletes vitamin D (Week 13). Same organ, three upstream causes.

Fracture healing (indirect / secondary)

1. **Hematoma formation** — bleeding at the site; the very first step.
2. **Soft (fibrocartilage) callus** bridges the gap.
3. **Hard (bony) callus** replaces it.
4. **Remodeling** restores normal shape and strength.

The fracture emergencies

Emergency	Mechanism	Clues
Compartment syndrome	Swelling/bleeding inside a closed fascial compartment → ↑ pressure → perfusion cut off → ischemia and necrosis	Pain out of proportion, unrelieved by analgesia ; tense swollen limb; paresthesia; the 6 P's (pain, pressure, paresthesia, pallor, pulselessness, paralysis — pulselessness is late). Treated with fasciotomy
Rhabdomyolysis	Muscle breakdown releases myoglobin , which precipitates in renal tubules	Acute kidney injury , dark tea-colored urine, ↑ creatine kinase, hyperkalemia → dysrhythmia
Fat embolism	Marrow fat from long-bone fractures enters circulation	Dyspnea, confusion, petechial rash — typically 12–72 hours after injury
Avascular necrosis / nonunion	Blood supply disrupted (classically femoral neck, scaphoid)	Persistent pain, failure to heal

RED FLAG — THE PHRASE TO RECOGNIZE

"**Pain out of proportion to the injury, unrelieved by opioids**" is the exam's signal for **compartment syndrome**. Do **not** elevate the limb above heart level or apply ice (both further reduce perfusion), and loosen restrictive casts/dressings. Waiting for pulselessness means waiting too long.

CROSS-WEEK THREAD Compartment syndrome → muscle necrosis → **rhabdomyolysis** → myoglobin → **intrarenal AKI** (Week 11), while released potassium causes **cardiac dysrhythmias** (Week 9). One limb injury, three systems.

PRACTICE

Q1. Which change explains vertebral compression fractures in osteoporosis?

- A. Reduced bone mass weakens vertebrae so they collapse under normal load
- B. Excessive bone formation
- C. Defective collagen present from birth
- D. Inadequate mineralization of newly formed bone only

A is correct. Osteoporosis reduces total bone mass, so porous vertebral bodies fail under everyday loads → height loss and kyphosis. Serum calcium is characteristically **normal**.

PRACTICE

Q2. Six hours after a tibial fracture a patient reports severe pain unrelieved by opioids with a tense, swollen calf and numbness. What is happening?

- A. Compartment syndrome — rising compartment pressure is cutting off perfusion
- B. Normal post-fracture inflammation
- C. Fat embolism syndrome
- D. Deep vein thrombosis

A is correct. Pressure in a closed fascial space exceeds capillary perfusion pressure, producing ischemia. **Pain out of proportion** is the earliest and most reliable sign; pulselessness is a late finding and waiting for it risks permanent damage.

PRACTICE

Q3. Which substance released in rhabdomyolysis causes acute kidney injury?

- A. Myoglobin, which precipitates in and obstructs renal tubules
- B. Creatinine
- C. Albumin
- D. Bilirubin

A is correct. Myoglobin released from damaged muscle is filtered and precipitates in tubules, causing obstruction and direct tubular toxicity. Released potassium simultaneously threatens cardiac rhythm.

	OSTEOARTHRITIS	RHEUMATOID ARTHRITIS	GOUT
Nature	Degenerative — mechanical wear	Autoimmune, systemic inflammation	Metabolic — crystal deposition
Lesion	Articular cartilage erodes ; bone spurs (osteophytes) form	Synovial inflammation → pannus invades and destroys cartilage and bone	Monosodium urate crystals deposit in the joint → acute inflammation
Pattern	Asymmetric ; weight-bearing joints (hips, knees, spine) and DIP joints	Symmetric ; small joints first (MCP, PIP , wrists) — and it sparcs the DIP	Monoarticular , classically the first MTP (podagra)
Morning stiffness	< 30 minutes ; worse after activity, better with rest	> 1 hour ; better with movement	Abrupt severe attacks, often at night
Systemic features	None	Yes — fatigue, low-grade fever, weight loss, rheumatoid nodules	Usually none between attacks; tophi in chronic disease
Signature findings	Heberden nodes (DIP), Bouchard nodes (PIP) , crepitus	Ulnar deviation , swan-neck and boutonnière deformities; RF and anti-CCP positive	Needle-shaped, negatively birefringent crystals; ↑ uric acid

MNEMONIC OA = "Old and Asymmetric, better with rest." RA = "Rheumatoid is Rise-and-stiff (>1 hr), symmetric, systemic." If the stem mentions **symmetry + prolonged morning stiffness + fatigue**, it is RA every time.

Pearl — the stiffness rule. **Inflammatory** arthritis is worse after **rest** and improves with **use** (RA: gelling overnight, loosens through the morning). **Degenerative** arthritis is worse with **use** and improves with **rest** (OA: hurts by the end of the day). One question about timing sorts the two.

Key idea — gout is a solubility problem. Hyperuricemia (overproduction from high purine intake, cell turnover, or underexcretion — often renal impairment or diuretics) → urate exceeds solubility → **crystals precipitate in cooler peripheral joints** → neutrophils attack them → intense inflammation. That is why the great toe, the coolest and most distal joint, is the classic site.

RED FLAG — SEPTIC ARTHRITIS

A **single hot, swollen, exquisitely painful joint with fever** must be assumed infected until arthrocentesis proves otherwise. Bacterial infection destroys cartilage within days — this is the "which arthritis is an emergency" answer.

PRACTICE

Q1. Which presentation is most consistent with rheumatoid arthritis?

- A. Symmetric small-joint pain in the hands with morning stiffness lasting two hours, plus fatigue and low-grade fever
- B. Asymmetric knee pain worse at the end of the day, relieved by rest
- C. Sudden excruciating pain in one great toe overnight
- D. A single hot swollen knee with high fever

A is correct. Symmetry, prolonged morning stiffness, small-joint involvement, and systemic symptoms define RA. Option B is OA, C is gout, and D is septic arthritis.

PRACTICE

Q2. Which mechanism causes joint destruction in rheumatoid arthritis?

- A. Autoimmune synovial inflammation forms pannus that erodes cartilage and bone
- B. Mechanical wear of articular cartilage over time
- C. Deposition of monosodium urate crystals
- D. Bacterial invasion of the joint space

A is correct. The inflamed, proliferating synovium (pannus) actively invades and destroys the joint — which is why RA is systemic and progressive rather than simply degenerative.

PRACTICE

Q3. Why does gout characteristically affect the first metatarsophalangeal joint?

- A. Urate is less soluble at the lower temperatures of distal peripheral joints, so crystals precipitate there
- B. That joint has the richest blood supply
- C. That joint lacks synovial fluid
- D. Uric acid is produced locally in the toe

A is correct. Solubility falls as temperature falls. The coolest, most distal joint is where supersaturated urate crystallizes first — producing podagra.

Disorder	Mechanism	Signature
Duchenne muscular dystrophy	X-linked recessive defect in dystrophin → progressive muscle fiber destruction replaced by fat and fibrous tissue	Boys age 3–5; Gower sign (walking the hands up the legs to stand), calf pseudohypertrophy , waddling gait, markedly ↑ creatine kinase; progressive with eventual cardiac and respiratory involvement
Myasthenia gravis	Autoimmune antibodies against acetylcholine receptors at the neuromuscular junction	Weakness that worsens with use and improves with rest; ptosis and diplopia first; worse at the end of the day; myasthenic crisis threatens respiratory muscles
Fibromyalgia	Central pain amplification (no inflammation or tissue destruction)	Widespread pain, fatigue, unrefreshing sleep, cognitive fog; normal labs and imaging
Strain vs sprain	Strain = muscle/tendon; sprain = ligament	Both: pain, swelling, limited motion (RICE management)
Herniated intervertebral disc	Nucleus pulposus protrudes through the annulus and compresses a nerve root	Lumbar: sciatica — radiating leg pain, numbness, weakness, worse with sitting/coughing/straining
Spinal stenosis	Narrowed canal compresses neural structures	Neurogenic claudication — leg pain with walking, relieved by sitting or leaning forward (flexion opens the canal)
Scoliosis	Lateral curvature with rotation	Adolescent screening: rib hump on forward bend , uneven shoulders/scapulae; severe curves impair pulmonary function
Osteoarthritic hip fracture	Fragility fracture on osteoporotic bone	Shortened, externally rotated leg; high morbidity in older adults
Plantar fasciitis	Inflammation of the plantar fascia at the calcaneal insertion	Heel pain worst with the first steps in the morning , easing with movement

Pearl — two claudications, opposite relief. **Vascular** claudication (PAD, Week 9) is relieved by **stopping**, regardless of position. **Neurogenic** claudication (spinal stenosis) is relieved by **flexing forward** — patients report they can push a shopping cart much farther than they can walk upright.

Pearl — the fatigability rule. Weakness that **worsens with repeated use and recovers with rest** points to the **neuromuscular junction** (myasthenia gravis). Weakness that is constant and progressive points to the **muscle fiber itself** (muscular dystrophy).

PRACTICE

Q1. A 4-year-old boy pushes his hands against his thighs to rise from the floor and has enlarged calves. Which mechanism explains this?

- A. Absent dystrophin causes muscle fiber degeneration with replacement by fat and connective tissue
- B. Antibodies block acetylcholine receptors
- C. Central amplification of pain signals
- D. Compression of a lumbar nerve root

A is correct. This is the **Gower sign** of Duchenne muscular dystrophy. Proximal weakness forces the child to "walk up" his own legs, and the enlarged calves are **pseudohypertrophy** — fat and fibrous tissue, not muscle.

PRACTICE

Q2. Which pattern is characteristic of myasthenia gravis?

- A. Weakness that worsens with sustained activity and improves after rest, with early ptosis and diplopia
- B. Constant progressive weakness beginning in the calves
- C. Widespread pain with normal strength
- D. Weakness relieved by exercise and worsened by rest

A is correct. Antibodies reduce available acetylcholine receptors, so repeated stimulation depletes transmission. The small, frequently used extraocular and eyelid muscles fatigue first.

PRACTICE

Q3. A patient's leg pain with walking is relieved by leaning forward over a shopping cart. Which condition does this suggest?

- A. Lumbar spinal stenosis — flexion widens the spinal canal and relieves neural compression
- B. Peripheral arterial disease
- C. Deep vein thrombosis
- D. Plantar fasciitis

A is correct. Position-dependent relief indicates a **mechanical** canal problem. Arterial claudication is relieved by rest alone and is unaffected by posture.

Condition	Age	Signature
Developmental dysplasia of the hip	Newborn/infant	Asymmetric thigh/gluteal folds, limited abduction, apparent leg-length difference; Ortolani (clunk on reduction) and Barlow (dislocatable) maneuvers
Clubfoot (talipes equinovarus)	Congenital	Foot turned inward and downward, not passively correctable; serial casting
Legg-Calvé-Perthes disease	~4–8 years	Avascular necrosis of the femoral head → painless limp or hip/knee/thigh pain, limited internal rotation
Slipped capital femoral epiphysis	Adolescent, often obese	Femoral head slips at the growth plate → hip or referred knee pain , limp, externally rotated leg — requires prompt surgical fixation
Osgood-Schlatter disease	Active adolescent	Traction injury at the tibial tubercle → anterior knee pain and a tender bump, worse with jumping/running
Growth plate (Salter-Harris) fracture	Any child	The physis is the weakest point in growing bone — injury there can cause growth arrest or limb-length discrepancy
Greenstick fracture	Young child	Bone bends and cracks incompletely — pediatric bone is more porous and flexible than adult bone
Juvenile idiopathic arthritis	Childhood	Persistent joint swelling >6 weeks, morning stiffness; systemic form has fever and rash; uveitis risk requires eye screening

Key idea — children's bones are different. They are **more porous and flexible** (greenstick and buckle fractures instead of complete breaks), they have an **open growth plate** that is weaker than surrounding ligament (so children fracture where adults sprain), and they **remodel far better** — but growth-plate injury risks permanent deformity.

RED FLAG — KNEE PAIN THAT IS ACTUALLY HIP DISEASE

Both **Legg-Calvé-Perthes** and **SCFE** commonly present with **referred knee or thigh pain**. A limping child complaining of knee pain needs the **hip** examined. Missing SCFE risks avascular necrosis of the femoral head.

PRACTICE

Q1. An obese 13-year-old has a limp and knee pain; the affected leg is externally rotated. Which condition should be suspected?

- A. Slipped capital femoral epiphysis — the femoral head displaces at the growth plate and refers pain to the knee
- B. Osgood-Schlatter disease
- C. Developmental dysplasia of the hip
- D. Juvenile idiopathic arthritis

A is correct. SCFE occurs in adolescents during rapid growth and is associated with obesity. Because hip pathology refers to the knee, evaluating only the knee misses the diagnosis — and delay risks avascular necrosis.

PRACTICE

Q2. Why is a growth-plate fracture of particular concern in children?

- A. Damage to the physis can arrest growth and cause limb-length discrepancy or angular deformity
- B. Growth plates cannot be visualized on imaging
- C. Children's bones do not remodel
- D. The growth plate has no blood supply

A is correct. The physis is where longitudinal growth occurs and is the weakest zone in a growing bone. Injuring it can permanently alter the length or alignment of the limb.

PRACTICE

Q3. Which finding suggests developmental dysplasia of the hip in an infant?

- A. Asymmetric thigh skinfolds with limited hip abduction and a positive Ortolani maneuver
- B. Anterior knee pain with a tender tibial tubercle
- C. Foot turned inward and downward at birth
- D. Painless limp in a 6-year-old with limited internal rotation

A is correct. Ortolani (clunk as the hip reduces) and Barlow (hip dislocates) plus asymmetric folds and limited abduction are the newborn screening findings. Option D describes Legg-Calvé-Perthes disease.

Key idea — skin is an organ with jobs. Barrier against microbes, fluid regulation, thermoregulation, sensation, and vitamin D synthesis. **Every major skin injury is a failure of those functions** — which is exactly why burns cause infection, hypovolemia, and hypothermia.

Burns — classified by depth

Depth	Layers involved	Appearance	Pain	Healing
Superficial (1st degree)	Epidermis only	Red, dry, no blisters (sunburn)	Painful	Days, no scarring
Partial thickness (2nd degree)	Epidermis + part of dermis	Blisters , moist, red, weeping	Very painful — nerve endings exposed	Weeks; may scar
Full thickness (3rd degree)	Entire dermis ± deeper	White, leathery, waxy, or charred; dry	Painless in the burned area — nerve endings destroyed	Requires grafting

Pearl — the painless burn is the worst burn. A full-thickness burn that does not hurt has destroyed the sensory nerve endings. Students frequently mis-rank severity by pain; on the exam, **less pain with a leathery white appearance = deeper injury**.

Key idea — the two phases of burn danger. First 24–48 hours: fluid. Massive capillary permeability causes plasma to shift into tissue → **hypovolemic (burn) shock**, hemoconcentration, and **hyperkalemia** from cell lysis. **After that: infection.** The barrier is gone and necrotic tissue is a culture medium — **sepsis becomes the leading cause of death**. Also watch for **hypothermia** (no thermoregulation) and, with facial/inhalation burns, **airway edema**.

RED FLAG — INHALATION INJURY

Facial burns, singed nasal hair, soot in the mouth, hoarseness, or stridor signal airway involvement. Progressive edema can obstruct the airway — airway assessment precedes fluid calculations. The **rule of nines** estimates total body surface area for resuscitation planning.

Pressure injuries

Unrelieved pressure over a bony prominence → capillary flow occluded → ischemia → tissue death. **Shear, friction, moisture, malnutrition, immobility, and impaired sensation** accelerate it.

Stage	Findings
Stage 1	Intact skin with non-blanchable erythema
Stage 2	Partial-thickness loss — shallow open ulcer or intact/ruptured blister; dermis exposed
Stage 3	Full-thickness loss — subcutaneous fat visible ; may have undermining/tunneling; bone/tendon/muscle not exposed
Stage 4	Full-thickness loss with exposed bone, tendon, or muscle
Unstageable	Base obscured by slough or eschar — depth cannot be determined until debrided
Deep tissue injury	Persistent non-blanchable deep red, maroon, or purple discoloration of intact skin

Pearl — blanching is the whole test in Stage 1. Redness that **blanches** (whitens with pressure, then refills) means perfusion is intact — reactive hyperemia. Redness that does **not** blanch means capillaries are already damaged. That single maneuver separates normal from injury.

CROSS-WEEK THREAD Everything that impaired wound healing in Week 9 impairs it here: diabetes, poor perfusion, low albumin (Week 11 nephrotic, Week 13 cirrhosis), steroids, smoking, and protein/vitamin C deficiency. Pressure injuries are wound healing plus sustained ischemia.

PRACTICE

Q1. A burn area appears white and leathery and the patient reports no pain there. What does this indicate?

- A. Full-thickness burn — sensory nerve endings in the dermis have been destroyed
- B. Superficial burn that is healing
- C. Partial-thickness burn with intact nerves
- D. Adequate pain management

A is correct. Absence of pain in a leathery, waxy-white burn reflects destruction of the entire dermis including nerve endings — the deepest and most serious injury, requiring grafting.

PRACTICE

Q2. During the first 24 hours after a major burn, which problem is the greatest threat?

- A. Hypovolemic shock from massive capillary leak and fluid shift out of the vasculature
- B. Wound infection and sepsis
- C. Contracture formation
- D. Hypokalemia

A is correct. Increased capillary permeability drives plasma into the interstitium, causing burn shock. Infection dominates *after* the first 48 hours, and cell lysis initially causes **hyper**kalemia, not hypokalemia.

PRACTICE

Q3. Intact skin over the sacrum shows redness that does not blanch with fingertip pressure. How is this classified?

- A. Stage 1 pressure injury
- B. Stage 2 pressure injury
- C. Unstageable pressure injury
- D. Normal reactive hyperemia

A is correct. Stage 1 is **intact skin with non-blanchable erythema**. If it blanched, perfusion would be intact and it would be normal reactive hyperemia. Stage 2 requires actual skin loss.

Inflammatory skin conditions

Condition	Mechanism	Signature
Atopic dermatitis (eczema)	Barrier dysfunction + type I hypersensitivity tendency	Itchy, dry, red patches; infants on face/scalp, children in flexural creases; part of the atopic triad with asthma and allergic rhinitis
Contact dermatitis	Type IV (delayed) hypersensitivity — e.g., poison ivy, nickel — or direct irritation	Rash in the distribution of contact , appearing 24–72 hours later
Psoriasis	Accelerated keratinocyte turnover driven by immune dysregulation	Thick silvery scales on erythematous plaques, extensor surfaces (elbows, knees), scalp; nail pitting; may have psoriatic arthritis
Urticaria (hives)	Histamine release → vascular leak into the dermis	Transient raised wheals with intense itch; angioedema with deeper involvement
Cellulitis	Bacterial infection of dermis and subcutaneous tissue (strep, staph)	Warm, red, swollen, tender, poorly demarcated area ± fever — spreading redness needs treatment
Necrotizing fasciitis	Rapidly spreading infection along fascial planes with tissue necrosis	Pain out of proportion , rapid progression, systemic toxicity, skin changes — surgical emergency

Pearl — flexor vs extensor. Eczema goes in the bends (antecubital and popliteal **flexural** creases). **Psoriasis goes on the points** (**extensor** elbows and knees). One word in the stem often decides the answer.

Pediatric and infectious rashes

Condition	Cause	Signature
Varicella (chickenpox)	Varicella-zoster virus	Itchy rash in all stages at once — macules, papules, vesicles, crusts; reactivates later as shingles in a dermatomal band
Measles (rubeola)	Measles virus	High fever, cough, coryza, conjunctivitis, then a rash spreading head to toe ; Koplik spots (white spots on buccal mucosa) precede it
Fifth disease	Parvovirus B19	"Slapped cheek" facial rash then a lacy rash on the body
Roseola	Human herpesvirus 6	High fever for days, then rash appears as the fever breaks
Scarlet fever	Group A streptococcus toxin	Sandpaper-textured rash, strawberry tongue , sore throat — treat to prevent rheumatic fever and post-strep glomerulonephritis (Week 11)
Impetigo	Staph/strep, superficial	Honey-colored crusts , highly contagious, common around the nose and mouth
Hand-foot-and-mouth	Coxsackievirus	Oral ulcers with vesicles on palms and soles
Tinea	Dermatophyte fungus	Annular (ring-shaped) scaly patch with central clearing
Scabies	Mite burrows in skin	Intense itching, worse at night ; burrows in web spaces and wrists

Skin cancer

Type	Behavior	Appearance
Basal cell carcinoma	Most common ; locally invasive, rarely metastasizes	Pearly, waxy papule with rolled borders and telangiectasias; may ulcerate. Sun-exposed areas
Squamous cell carcinoma	Can metastasize ; arises from actinic keratosis	Scaly, crusted, firm red nodule or non-healing ulcer
Melanoma	Most deadly — metastasizes early via lymph and blood	ABCDE: Asymmetry, Border irregularity, Color variation, Diameter >6 mm, Evolving/changing

RED FLAG — "EVOLVING" IS THE STRONGEST SINGLE CLUE

A mole that is **changing** in size, shape, color, or symptom (itching, bleeding) warrants evaluation regardless of how it scores on the other letters. UV exposure, fair skin, many or atypical moles, and family history raise risk.

PRACTICE

Q1. Which mechanism produces the thick silvery scaling plaques of psoriasis?

- A. Immune-driven acceleration of keratinocyte proliferation and turnover
- B. Bacterial infection of the dermis
- C. Histamine release causing transient wheals
- D. Delayed hypersensitivity to an external contactant

A is correct. Keratinocytes mature and shed far faster than normal, so immature cells pile up as thick silvery scale over inflamed plaques — typically on extensor surfaces.

PRACTICE

Q2. Which lesion description is most concerning for melanoma?

- A. An asymmetric mole with irregular borders, multiple colors, and recent change in size
- B. A pearly papule with rolled borders and visible small vessels
- C. A scaly crusted nodule on the lower lip
- D. An annular scaly patch with central clearing

A is correct. That describes the ABCDE criteria. Option B is basal cell carcinoma, C is squamous cell carcinoma, and D is tinea (a fungal infection, not a cancer).

PRACTICE

Q3. A child has a high fever for three days; as the fever resolves, a rash appears. Which illness is most likely?

- A. Roseola
- B. Measles
- C. Scarlet fever
- D. Fifth disease

A is correct. Roseola's signature is a **rash that appears as the fever breaks**. In measles the rash appears while the child is still febrile and ill, preceded by Koplik spots.

PUT IT TOGETHER

Cross-System Threads & Confusion Pairs

Final exams reward students who see how the systems connect. These are the links and the look-alikes.

42 Ten Cross-System Threads

Each thread below is a single mechanism that shows up in three or more weeks. If a question feels unfamiliar, check whether it is one of these wearing a different costume.

1. The oxygen delivery chain

W10 → W8 → W9 Oxygen must be **loaded** (lungs), **carried** (hemoglobin), and **delivered** (cardiac output). Break any link and tissue is hypoxic. This is why a patient with COPD **and** anemia **and** heart failure decompensates far out of proportion to any single diagnosis — and why a normal SaO_2 never rules out tissue hypoxia.

2. Acid–base: lungs fast, kidneys slow

W10 ↔ W11 A metabolic acidosis (DKA, shock, CKD, diarrhea) is compensated in **minutes** by hyperventilation (**Kussmaul**). A respiratory acidosis is compensated over **days** by renal bicarbonate retention. **Full compensation therefore implies chronicity.**

3. Edema always has one of four causes

W9 · W11 · W13 · W14 **↑hydrostatic** (heart failure, portal hypertension) · **↓oncotic** (nephrotic syndrome, cirrhosis, malnutrition) · **↑permeability** (burns, sepsis, inflammation) · **lymphatic obstruction**. Every swelling question in Weeks 8–14 resolves to one of these four.

4. The kidney–blood axis

W11 → W8 Kidney makes **EPO**: CKD → **normocytic anemia**. Reverse it and chronic hypoxia (Week 10) drives EPO up → **secondary polycythemia**. Uremia also impairs **platelet function** → bleeding with a normal platelet count.

5. The liver is the coagulation factory

W13 → W8 Cirrhosis → no clotting factors (**↑INR**) + splenic sequestration (**↓platelets**) + varices (a bleeding source). Fat malabsorption → no **vitamin K** → **↑PT**. A "liver question" is usually also a coagulation question.

6. Vitamin D → calcium → bone

W11 · W13 → W14 The **kidney** activates vitamin D and the **gut** absorbs the calcium. CKD → **renal osteodystrophy**. Fat malabsorption (pancreatic insufficiency, biliary obstruction, celiac) →

osteomalacia. Menopause removes estrogen's brake → **osteoporosis**. Three routes to weak bone.

7. Virchow's triad → DVT → PE

W8 → **W9** → **W10** Stasis + endothelial injury + hypercoagulability creates a clot in the leg that lodges in the lung. A large PE becomes **obstructive shock**. This one triad spans three weeks and is a favorite exam chain.

8. RAAS: the same hormone system, three chapters

W11 ↔ **W9** The **kidney** senses low perfusion and activates RAAS, which is lifesaving in hemorrhage and **harmful in heart failure** (raises preload and afterload). It also drives hypertension, which then damages the heart, kidney, brain, and vessels.

9. Portal hypertension's downstream chaos

W13 → **W8** • **W9** Scarred liver → portal pressure ↑ → **varices** (hemorrhage), **ascites** (edema physiology), **splenomegaly** → **thrombocytopenia**, and encephalopathy from lost detoxification.

10. Diabetes is a multi-system disease

W9 • **W11** • **W12** • **W14** **Macrovascular**: CAD, PAD, stroke. **Microvascular**: nephropathy (a leading cause of CKD), retinopathy, neuropathy. **Immune**: impaired leukocyte function → infection and poor wound healing. **Autonomic**: silent MI, erectile dysfunction, gastroparesis. If a stem mentions diabetes, it is telling you why the primary problem is worse.

PRACTICE

Q1. A patient with cirrhosis has an INR of 2.3 and a platelet count of 62,000. Which two mechanisms explain these findings?

- A. Reduced hepatic clotting factor synthesis and splenic platelet sequestration from portal hypertension
- B. Vitamin B₁₂ deficiency and marrow failure
- C. Disseminated intravascular coagulation and hemolysis
- D. Renal failure and uremic toxin accumulation

A is correct. The failing hepatocytes cannot synthesize factors (↑INR), while portal hypertension congests the spleen, which traps platelets (↓count). Two different consequences of the same cirrhosis.

PRACTICE

Q2. A patient with advanced COPD has hematocrit 57% and right-sided heart failure. Which single mechanism explains both?

- A. Chronic hypoxemia — it stimulates erythropoietin and causes pulmonary vasoconstriction leading to cor pulmonale
- B. Chronic carbon dioxide retention alone
- C. Left ventricular systolic failure
- D. Iron overload from repeated transfusion

A is correct. Low oxygen drives renal EPO release (polycythemia) and simultaneously constricts pulmonary vessels, raising right ventricular afterload until it fails — cor pulmonale. One stimulus, two organ consequences.

PRACTICE

Q3. A patient with a crush injury develops dark urine, potassium 6.4 mEq/L, and rising creatinine. Trace the mechanism.

- A. Muscle necrosis releases myoglobin and potassium; myoglobin obstructs renal tubules causing intrarenal AKI while hyperkalemia threatens cardiac rhythm
- B. Hemolysis releases hemoglobin causing prerenal failure
- C. Obstruction of the ureters causes postrenal failure
- D. Immune complex deposition causes glomerulonephritis

A is correct. This is the Week 14 → Week 11 → Week 9 chain: rhabdomyolysis → myoglobinuric acute tubular necrosis → and the released intracellular potassium creates an immediate dysrhythmia risk.

43 Confusion Pairs — A vs B

Most missed questions are not "I never learned this" — they are "I mixed up two things that look alike." Here is the discriminator for each.

Pair	The one clue that separates them
B₁₂ vs folate deficiency	Neurologic signs → B ₁₂
Iron deficiency vs anemia of chronic disease	Ferritin low → iron deficiency. Ferritin high → chronic disease
Platelet vs factor bleeding	Petechiae/mucosal, immediate → platelet. Deep/joint, delayed → factor
Primary vs secondary polycythemia	EPO low → primary (vera). EPO high → secondary (hypoxia)
TACO vs TRALI	Hypertension + JVD → TACO (volume). Hypoxemia with normal/low BP → TRALI (lung injury)
Left vs right heart failure	Crackles/orthopnea → left. JVD/peripheral edema/hepatomegaly → right
Systolic vs diastolic failure	EF reduced → systolic. EF preserved → diastolic (filling problem)
Unstable angina vs MI	Troponin — elevated only in MI
Arterial vs venous ulcer	Dangling relieves + absent pulses + toe ulcer → arterial. Elevation relieves + brown staining + medial malleolus → venous
Obstructive vs restrictive lung disease	FEV₁/FVC reduced → obstructive. Ratio normal, all volumes down → restrictive
Shunt vs dead space	Refractory to O₂ → shunt (pneumonia). Ventilated, not perfused → dead space (PE)
Pneumothorax vs pleural effusion	Hyperresonant → air. Dull → fluid
ARDS vs cardiogenic pulmonary edema	Normal filling pressures, refractory hypoxemia → ARDS. JVD, S3, ↑BNP, diuresis helps → cardiogenic
Croup vs epiglottitis	Barking cough, gradual → croup. Drooling, no cough, abrupt → epiglottitis
Prerenal vs intrarenal AKI	BUN:Cr >20:1, low urine Na → prerenal. Muddy brown casts, high urine Na → ATN

Nephritic vs nephrotic	Hematuria + RBC casts + HTN → nephritic. Proteinuria >3.5 g + low albumin + severe edema → nephrotic
Cystitis vs pyelonephritis	Fever + CVA tenderness + WBC casts → pyelonephritis
AKI vs CKD	Anemia + small kidneys + bone disease → chronic
Torsion vs epididymitis	Absent cremasteric reflex, elevation doesn't help → torsion (surgery now)
BPH vs prostate cancer	Smooth, symmetric, obstructive symptoms early → BPH. Hard irregular nodule, silent early, bone pain → cancer
Gastric vs duodenal ulcer	Pain worse with food → gastric. Relieved by food, returns 2–3 h later → duodenal
Crohn vs ulcerative colitis	Skip lesions, transmural, fistulas, anywhere → Crohn. Continuous, mucosal, bloody, rectum-up → UC
IBD vs IBS	Bleeding, weight loss, fever, abnormal labs → IBD (inflammatory). None of those → IBS (functional)
Prehepatic vs posthepatic jaundice	Clay stool + dark urine + pruritus → obstructive (posthepatic)
Vomiting vs diarrhea (acid–base)	Vomiting loses acid → alkalosis. Diarrhea loses bicarbonate → acidosis
Pyloric stenosis vs intussusception	Projectile non-bilious vomiting, olive, 2–8 wk → pyloric. Currant-jelly stool, sausage mass, 3 mo–3 yr → intussusception
Osteoporosis vs osteomalacia	Normal calcium, too little bone → osteoporosis. Vitamin D deficiency, soft bone → osteomalacia
OA vs RA	Asymmetric, <30 min stiffness, worse with use → OA. Symmetric, >1 h stiffness, systemic → RA
Vascular vs neurogenic claudication	Relieved by stopping → vascular. Relieved by leaning forward → spinal stenosis
Eczema vs psoriasis	Flexural, itchy, dry → eczema. Extensor, silvery thick scale → psoriasis
Partial vs full-thickness burn	Blisters, very painful → partial. Leathery white, painless → full thickness

Roseola vs measles

Rash appears as fever breaks → roseola. Rash with ongoing fever + Koplik spots → measles

★ CHEAT SHEETS

Master Reference

The tables worth a final glance the morning of the exam.

Lab signatures — read the arrows, name the disorder

Disorder	Platelets	PT/INR	aPTT	Other
DIC	↓↓	↑	↑	↓fibrinogen, ↑↑D-dimer
Hemophilia A	n	n	↑	Factor VIII deficiency; hemarthrosis
ITP	↓	n	n	Petechiae; often post-viral in children
Liver failure / vitamin K deficiency	n or ↓	↑	n → ↑	↓albumin, ↑bilirubin, ↑ammonia
von Willebrand disease	n	n	n or ↑	Mucosal bleeding, heavy menses

Anemia	MCV	Retic	Discriminator
Iron deficiency	↓ Micro	↓	↓ferritin, ↑TIBC
Anemia of chronic disease	Micro or normal	↓	↑ferritin, ↓TIBC
Thalassemia	↓ Micro	Variable	Normal iron studies
B ₁₂ deficiency	↑ Macro	↓	Neurologic signs
Folate deficiency	↑ Macro	↓	No neurologic signs
CKD	Normal	↓	↓EPO, ↑creatinine
Hemolytic	Normal	↑↑	↑LDH, ↑indirect bili, ↓haptoglobin
Aplastic	Normal	↓↓	Pancytopenia

ABG quick table

pH	PaCO ₂	HCO ₃	Interpretation
↓	↑	n	Uncompensated respiratory acidosis (acute)
↓	↑	↑	Partially compensated respiratory acidosis
n (<7.40)	↑	↑	Fully compensated respiratory acidosis (chronic — think COPD)
↑	↓	n	Uncompensated respiratory alkalosis (anxiety, pain, fever, early PE)
↓	n	↓	Uncompensated metabolic acidosis (DKA, shock, renal failure, diarrhea)
↓	↓	↓	Partially compensated metabolic acidosis (Kussmaul breathing)
↑	n	↑	Uncompensated metabolic alkalosis (vomiting, NG suction, diuretics)

REMEMBER ROME — Respiratory Opposite, Metabolic Equal. And: **the value that matches the pH direction is the problem; the one that opposes it is the compensation.**

Don't-miss emergencies (Weeks 8–14)

Emergency	The tell	Why it's urgent
Acute hemolytic transfusion reaction	Fever, flank pain, dark urine during transfusion	Stop transfusion — progresses to AKI and DIC
DIC	Oozing from every site + ↓platelets, ↑PT/aPTT, ↑D-dimer	Treat the trigger (usually sepsis)
Fever in sickle cell disease	Any fever in a functionally asplenic patient	Overwhelming encapsulated-organism sepsis
Myocardial infarction	Chest pain >20 min + ↑troponin (or atypical in women/diabetes/elderly)	Time is muscle; dysrhythmia kills early
Tension pneumothorax	Tracheal deviation away , JVD, absent breath sounds, hypotension	Obstructive shock — decompress immediately
Pulmonary embolism	Sudden dyspnea + pleuritic pain + hypoxemia with low CO ₂	Can cause obstructive shock and sudden death
Silent chest in asthma	Wheeze disappears, CO ₂ normalizing, drowsiness	Impending respiratory arrest
Epiglottitis	Drooling, dysphagia, dysphonia, distress — no cough	Do not examine the throat; total obstruction risk
Hyperkalemia in AKI	Oliguria + K ⁺ rising, peaked T waves	Cardiac arrest
Testicular torsion	Sudden pain, high-riding testis, absent cremasteric reflex	~6-hour salvage window
Ruptured ectopic pregnancy	Unilateral pelvic pain + missed period + instability	Rapid hemorrhagic shock
Postmenopausal bleeding	Any bleeding after menopause	Endometrial carcinoma until excluded
Variceal hemorrhage	Hematemesis in a cirrhotic patient	Bleeding source + no factors + no platelets

Perforated ulcer / peritonitis	Sudden pain, rigid board-like abdomen , rebound	Surgical emergency
Bowel strangulation	Obstruction + fever, tachycardia, constant pain	Ischemia → necrosis → perforation
Compartment syndrome	Pain out of proportion, unrelieved by opioids	Fasciotomy; pulselessness is a late sign
Septic arthritis	Single hot swollen joint + fever	Cartilage destroyed within days
Necrotizing fasciitis	Pain out of proportion + rapid spread + toxicity	Surgical debridement
Inhalation injury with burns	Facial burns, singed nares, hoarseness, stridor	Airway edema → obstruction
Wilms tumor	Unilateral abdominal mass in a 2–5 year old	Do not palpate — seeding risk

Pediatric quick reference by age

Age	Condition	Signature
Preterm newborn	Neonatal RDS; NEC; PDA	Surfactant deficiency; bloody stools + distension; machinery murmur
First 48 h	Hirschsprung	No meconium passed
>2 weeks jaundice	Biliary atresia	Clay stools, dark urine
2–8 weeks	Pyloric stenosis	Projectile non-bilious vomiting, olive, metabolic alkalosis
2–3 months	Physiologic anemia of infancy	Normal nadir, self-limited
<2 years	Bronchiolitis (RSV)	Wheeze, retractions, apnea risk
3 mo–3 yr	Intussusception	Currant-jelly stool, sausage mass, legs drawn up
6 mo–3 yr	Croup	Barking cough, stridor
2–5 yr	Wilms tumor; HUS	Abdominal mass (don't palpate); anemia + ↓platelets + AKI after bloody diarrhea
3–5 yr	Duchenne MD	Gower sign, calf pseudohypertrophy
4–8 yr	Legg-Calvé-Perthes	Painless limp, referred knee pain
School age	Post-strep GN; minimal change disease	Cola urine + HTN; frothy urine + periorbital edema
Adolescent	SCFE; Osgood-Schlatter; testicular torsion; scoliosis	Obese teen with knee pain; tender tibial tubercle; acute scrotum; rib hump
15–35 yr	Testicular cancer	Painless firm mass, no transillumination



40-Question Cumulative Practice Exam

Mixed order across Weeks 8–14 — the way the real exam will feel. Answer all 40 before revealing anything.

Q1 • W8

A patient's labs show MCV 105 fL with numbness and an unsteady gait. Which deficiency is most likely?

- A. Vitamin B₁₂
- B. Folate
- C. Iron
- D. Erythropoietin

A. Macrocytosis plus neurologic findings is B₁₂; folate causes macrocytosis without neurologic deficits.

Q2 • W9

Which finding indicates left-sided rather than right-sided heart failure?

- A. Crackles and orthopnea
- B. Jugular venous distention
- C. Hepatomegaly
- D. Dependent peripheral edema

A. Left failure backs blood into the lungs. B, C, and D are systemic venous congestion from right failure.

Q3 · W10

ABG: pH 7.50, PaCO₂ 28, HCO₃ 24. Which disorder is this?

- A. Uncompensated respiratory alkalosis
- B. Metabolic alkalosis
- C. Compensated metabolic acidosis
- D. Respiratory acidosis

A. High pH with low CO₂ (opposite directions = respiratory) and a normal bicarbonate = uncompensated respiratory alkalosis, typical of hyperventilation.

Q4 · W11

Which urinary finding localizes infection to the kidney rather than the bladder?

- A. White blood cell casts
- B. Bacteriuria
- C. Positive nitrites
- D. Cloudy urine

A. Casts form in renal tubules, so they can only originate in the kidney. The others occur in cystitis as well.

Q5 · W12

Which mechanism causes anovulation in polycystic ovary syndrome?

- A. Insulin resistance raises ovarian androgen production, arresting follicular development
- B. Excess progesterone suppresses the LH surge
- C. Premature depletion of all ovarian follicles
- D. Pituitary failure to produce FSH

A. Hyperinsulinemia drives androgen excess, follicles arrest before ovulating, and the result is anovulation with polycystic-appearing ovaries.

Q6 · W13

Which feature distinguishes ulcerative colitis from Crohn disease?

- A. Continuous mucosal inflammation beginning at the rectum
- B. Skip lesions with normal bowel between
- C. Transmural inflammation
- D. Fistula formation

A. UC is continuous, superficial, and rectum-upward. B, C, and D are all Crohn features — transmural depth is what permits fistulas.

Q7 · W14

Which finding is most concerning after a long-bone fracture?

- A. Severe pain unrelieved by opioids with a tense swollen limb
- B. Mild swelling that improves with elevation
- C. Bruising around the fracture site
- D. Low-grade discomfort with movement

A. Compartment syndrome. Pain out of proportion and unrelieved by analgesia is the earliest reliable sign; waiting for pulselessness is too late.

Q8 · W8

Which lab pattern confirms disseminated intravascular coagulation?

- A. Low platelets, prolonged PT and aPTT, low fibrinogen, elevated D-dimer
- B. Low platelets with normal PT and aPTT
- C. Prolonged aPTT with normal PT and platelets
- D. Normal platelets with prolonged PT only

A. All four numbers deranged with a high D-dimer proves simultaneous clot formation and breakdown. B is ITP, C is hemophilia A, D is early liver/vitamin K deficiency.

Q9 · W9

Which compensatory mechanism most directly worsens the workload of a failing heart?

- A. RAAS activation increasing sodium/water retention and vasoconstriction
- B. Natriuretic peptide release
- C. Increased 2,3-DPG
- D. Peripheral vasodilation

A. RAAS raises both preload and afterload. Natriuretic peptides are the one counter-regulatory (helpful) system.

Q10 · W10

Which condition produces hypoxemia that does not improve with supplemental oxygen?

- A. Lobar pneumonia causing shunt
- B. High-altitude exposure
- C. Opioid-induced hypoventilation
- D. Mild anemia

A. In a true shunt, blood bypasses ventilated alveoli entirely, so added oxygen never reaches it. The others respond to oxygen or ventilation.

Q11 · W11

A dehydrated patient has BUN:creatinine 30:1 with low urine sodium. Which AKI category is this?

- A. Prerenal
- B. Intrarenal
- C. Postrenal
- D. Chronic kidney disease

A. Intact tubules avidly conserving sodium and urea indicate a perfusion problem, not tissue damage.

Q12 · W12

Which finding requires immediate surgical evaluation?

- A. Sudden scrotal pain with an absent cremasteric reflex
- B. Gradual scrotal pain relieved by elevation with fever
- C. A soft scrotal swelling that transilluminates
- D. A "bag of worms" sensation on the left

A. Testicular torsion. B is epididymitis, C is hydrocele, D is varicocele — none are surgical emergencies.

Q13 · W13

A patient has jaundice, dark urine, and clay-colored stools. Which type of hyperbilirubinemia is present?

- A. Conjugated, from obstructed bile flow
- B. Unconjugated, from hemolysis
- C. Unconjugated, from impaired uptake
- D. Mixed, from viral hepatitis only

A. Pale stool means bile is not reaching the intestine; the water-soluble conjugated bilirubin backs up into blood and spills into urine.

Q14 · W14

Which pattern indicates rheumatoid arthritis rather than osteoarthritis?

- A. Symmetric small-joint involvement with morning stiffness over one hour
- B. Asymmetric knee pain worse after activity
- C. Heberden nodes at the DIP joints
- D. Crepitus with joint movement

A. Symmetry plus prolonged morning stiffness is the RA signature; RA characteristically spares the DIP joints.

Q15 · W8

Which trigger is most likely to precipitate a sickle cell vaso-occlusive crisis?

- A. Dehydration during a febrile illness
- B. Adequate hydration
- C. Warm ambient temperature
- D. Supplemental oxygen therapy

A. Dehydration concentrates HbS and infection promotes deoxygenation — both drive polymerization and sickling.

Q16 · W9

A leg ulcer at the medial malleolus with brown skin staining improves with elevation. What is the cause?

- A. Chronic venous insufficiency
- B. Peripheral arterial disease
- C. Diabetic neuropathic ulcer only
- D. Pressure injury

A. Elevation relief, hemosiderin staining, and medial malleolar location are venous. Arterial ulcers occur on toes/heels and improve with dangling.

Q17 · W10

Which spirometry result indicates obstructive lung disease?

- A. Reduced FEV_1/FVC ratio with increased residual volume
- B. Reduced total lung capacity with normal ratio
- C. Increased FEV_1/FVC ratio with reduced volumes
- D. Normal ratio with reduced compliance

A. Obstruction limits expiratory flow and traps air. Options B, C, and D describe restriction.

Q18 · W11

Which set of findings characterizes nephrotic syndrome?

- A. Proteinuria over 3.5 g/day, hypoalbuminemia, generalized edema, hyperlipidemia
- B. Hematuria with RBC casts and hypertension
- C. Fever with costovertebral angle tenderness
- D. Muddy brown casts with high urine sodium

A. That is the nephrotic tetrad. B is nephritic, C is pyelonephritis, D is acute tubular necrosis.

Q19 · W12

Why is early prostate cancer often asymptomatic while BPH causes urinary symptoms?

- A. Cancer typically arises peripherally while BPH arises centrally around the urethra
- B. Cancer grows more slowly than BPH in all cases
- C. Cancer does not enlarge the prostate
- D. BPH invades the bladder wall

A. Location determines early symptoms. Symptom severity is not a measure of seriousness here.

Q20 · W13

Which mechanism explains hepatic encephalopathy?

- A. Accumulation of ammonia the failing liver cannot convert to urea
- B. Excess conjugated bilirubin crossing the blood-brain barrier
- C. Deficiency of clotting factors
- D. Portal hypertension compressing cerebral vessels

A. Ammonia from protein/bacterial metabolism reaches the brain when detoxification fails — producing confusion and asterixis. A GI bleed worsens it by adding a protein load.

Q21 · W14

A pressure area shows intact skin with redness that does not blanch. Which stage is this?

- A. Stage 1
- B. Stage 2
- C. Stage 3
- D. Deep tissue injury

A. Stage 1 is intact skin with non-blanchable erythema. Stage 2 requires partial-thickness skin loss.

Q22 · W8

A transfusion is started; 15 minutes later the patient has fever, flank pain, and dark urine. What is the priority action?

- A. Stop the transfusion immediately and maintain IV access with normal saline
- B. Slow the transfusion rate and continue
- C. Administer an antipyretic and continue
- D. Elevate the head of the bed and continue

A. These findings indicate acute hemolytic reaction from ABO incompatibility, which can progress to AKI and DIC. Stopping the infusion is the first action.

Q23 · W9

Which congenital defect produces cyanosis?

- A. Tetralogy of Fallot
- B. Atrial septal defect
- C. Ventricular septal defect
- D. Patent ductus arteriosus

A. Tetralogy shunts right-to-left, sending deoxygenated blood into systemic circulation. B, C, and D are left-to-right (acyanotic) shunts.

Q24 · W10

Which finding indicates a tension pneumothorax?

- A. Tracheal deviation away from the affected side with hypotension and JVD
- B. Dullness to percussion over the affected side
- C. Bilateral crackles with pink frothy sputum
- D. Pleuritic pain with normal blood pressure

A. Mediastinal shift obstructing venous return is what makes it a tension — and obstructive shock. Dullness would suggest fluid, not air.

Q25 · W11

Which consequence of chronic kidney disease results from impaired vitamin D activation?

- A. Hypocalcemia with secondary hyperparathyroidism and bone demineralization
- B. Normocytic anemia
- C. Metabolic alkalosis
- D. Hypokalemia

A. Without calcitriol the gut cannot absorb calcium; PTH rises and pulls calcium from bone — renal osteodystrophy. Anemia is from EPO loss, and CKD causes acidosis and hyper-kalemia.

Q26 · W12

Which change explains accelerated bone loss after menopause?

- A. Loss of estrogen's inhibition of osteoclast activity
- B. Increased calcitonin secretion
- C. Reduced parathyroid hormone
- D. Increased osteoblast activity

A. Estrogen restrains bone resorption. Its withdrawal lets osteoclast activity outpace formation.

Q27 · W13

A 6-week-old has projectile non-bilious vomiting and remains hungry afterward. Which condition is this?

- A. Pyloric stenosis
- B. Intussusception
- C. Hirschsprung disease
- D. Biliary atresia

A. Non-bilious vomiting places the obstruction at or above the pylorus. The age, projectile pattern, persistent hunger, and olive-shaped mass are classic.

Q28 · W14

Which describes a full-thickness burn?

- A. White, leathery, dry, and painless in the burned area
- B. Red and dry with no blisters
- C. Moist with blisters and severe pain
- D. Superficial redness that blanches

A. Destruction of the full dermis eliminates nerve endings, so the deepest burn is the least painful in the burned area.

Q29 · W8+W11

A patient with CKD has hemoglobin 8.5 g/dL and normal MCV. Which treatment addresses the mechanism?

- A. Erythropoiesis-stimulating therapy, because the kidney no longer produces EPO
- B. Vitamin B₁₂ injections
- C. Folate supplementation
- D. Corticosteroids for immune hemolysis

A. The anemia is normocytic and caused by absent EPO signaling, so replacing the missing hormone addresses the actual defect.

Q30 · W9+W10

A postoperative immobile patient develops sudden dyspnea, pleuritic chest pain, and hypoxemia with a low PaCO₂. What happened?

- A. Pulmonary embolism from a deep vein thrombosis
- B. Acute myocardial infarction
- C. Pneumothorax
- D. Aspiration pneumonia

A. Virchow's triad (stasis, surgical endothelial injury, postoperative hypercoagulability) produced a DVT that embolized. Hyperventilation lowers the CO₂ while hypoxemia persists.

Q31 · W10

A 2-year-old has a barking cough and inspiratory stridor that began gradually at night. Which condition is this?

- A. Croup
- B. Epiglottitis
- C. Bronchiolitis
- D. Asthma

A. Croup barks; epiglottitis drools and has no cough. Bronchiolitis wheezes with copious secretions in infants under 2.

Q32 · W13

Which mechanism produces ascites in cirrhosis?

- A. Portal hypertension plus hypoalbuminemia
- B. Increased albumin synthesis
- C. Lymphatic obstruction from gallstones
- D. Excess pancreatic enzyme secretion

A. Hydrostatic pressure pushes fluid out while low oncotic pressure fails to pull it back — both Starling forces deranged in the same direction.

Q33 · W11

During the oliguric phase of AKI, which value requires the most urgent monitoring?

- A. Serum potassium
- B. Serum albumin
- C. Serum calcium
- D. Hemoglobin A1c

A. Potassium accumulates when excretion stops and causes lethal dysrhythmias — the most immediate threat in oliguria.

Q34 · W14

Which mole feature is most concerning for melanoma?

- A. Recent change in size and color with irregular borders
- B. Uniform brown color with regular borders
- C. Diameter of 3 mm
- D. Symmetry on both halves

A. Asymmetry, border irregularity, color variation, diameter over 6 mm, and evolution are the ABCDE criteria — "evolving" is the strongest single clue.

Q35 · W9

Which laboratory result distinguishes myocardial infarction from unstable angina?

- A. Elevated troponin
- B. Elevated BNP
- C. Elevated white blood cell count
- D. Elevated D-dimer

A. Troponin leaks only when myocytes die. BNP reflects ventricular stretch, and D-dimer reflects clot breakdown.

Q36 · W12

Which long-term risk is increased by chronic anovulation?

- A. Endometrial hyperplasia and carcinoma from unopposed estrogen
- B. Osteoporosis from excess progesterone
- C. Cervical cancer from progesterone deficiency
- D. Ovarian torsion

A. No ovulation means no corpus luteum and no progesterone, so estrogen stimulates the endometrium continuously without opposition.

Q37 · W13

A patient with a bowel obstruction becomes hypotensive and tachycardic. Which mechanism explains this?

- A. Fluid sequestration into the bowel lumen and wall plus vomiting losses
- B. Massive gastrointestinal hemorrhage
- C. Septic vasodilation from bacteremia
- D. Cardiogenic pump failure

A. Third-spacing removes large volumes from the circulation without visible loss, producing hypovolemia and electrolyte derangement.

Q38 · W8

Which finding indicates hemolysis rather than decreased red cell production?

- A. Elevated reticulocyte count with elevated LDH and low haptoglobin
- B. Low reticulocyte count with low ferritin
- C. Pancytopenia with a hypocellular marrow
- D. Normal reticulocyte count with elevated creatinine

A. A high reticulocyte count proves the marrow is responding appropriately; the LDH, bilirubin, and haptoglobin changes confirm cells are being destroyed.

Q39 · W14 + W11

A crush-injury patient has tea-colored urine and rising creatinine. Which mechanism is responsible?

- A. Myoglobin from damaged muscle precipitating in renal tubules
- B. Bilirubin from hemolysis
- C. Glomerular immune complex deposition
- D. Ureteral obstruction

A. Rhabdomyolysis releases myoglobin, which obstructs and injures tubules — an intrarenal AKI. Released potassium adds cardiac risk.

Q40 · W10 + W9 + W8

A patient with severe COPD has hematocrit 58%, JVD, and peripheral edema with a normal left ventricular ejection fraction. What is the explanation?

- A. Chronic hypoxemia caused secondary polycythemia and pulmonary hypertension leading to cor pulmonale
- B. Left-sided systolic heart failure
- C. Nephrotic syndrome with hypoalbuminemia
- D. Polycythemia vera with independent right heart disease

A. One stimulus — chronic hypoxemia — raises EPO (polycythemia) and constricts pulmonary vessels, raising right ventricular afterload until it fails. The normal left EF rules out left-sided failure as the cause.



Seven-Day Study Plan

If you have a week, use it in this order — heaviest and most connected material first.

Day 1 — Week 8

Day 2 — Week 9

Day 3 — Week 10

Blood. Build the four-number coagulation table from memory. Drill the MCV framework and the hemolysis signature. Do topics 1–6.

Heart. Draw left vs right failure from scratch. Write out the compensation table and why each becomes harmful. Topics 7–12.

Lungs. Do 20 ABGs using the 4-step method until it is automatic. Obstructive vs restrictive, shunt vs dead space. Topics 13–18.

Day 4 — Week 11

Kidney. Memorize the pre/intra/postrenal table and the six CKD consequences. Nephritic vs nephrotic. Topics 19–24.

Day 5 — Weeks 12 & 13

Reproductive + GI. Focus on PCOS, BPH vs cancer, the two urologic emergencies, Crohn vs UC, and the cirrhosis cascade. Topics 25–35.

Day 6 — Week 14 + integration

MSK/skin: the arthritis trio, fracture emergencies, burns, pressure stages, rashes. Then read topics 42–43 (threads and confusion pairs) twice.

Day 7 — Rehearsal

Take the 40-question exam cold, timed, no notes. Score it. Re-study only the topics behind your misses, then re-read the confusion-pairs table and the emergencies cheat sheet.

THREE HABITS THAT RAISE SCORES ON THIS EXAM

1. **Answer the mechanism, not the disease.** When a stem asks "which change explains," the correct option is a *process* (increased pressure, impaired mineralization, consumed factors), not a diagnosis.
2. **Find the discriminator before reading the options.** Decide what single fact separates the likely candidates, then look for the option containing it. This prevents being talked into a plausible-sounding distractor.
3. **Check whether the question is asking "what is happening" or "what do I do first."** Priority questions almost always want **stop the harm** (stop the transfusion, decompress the chest, relieve the pressure) before assessment or documentation.

